

UPDATE:



Appendix B
Demographics & Land Use

APPENDIX B — Demographics & Land Use

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Population & Employment Forecasts

DVRPC is required to maintain population and employment forecasts with a horizon of at least 20 years, or to the Plan's horizon year. The 2050 Version 2.1 (2050 v2.1) Population and Employment Forecast updates the 2050 v1.0 forecasts adopted in 2021 with more recent data from the 2020 Decennial Census, U.S. Bureau of Economic Analysis (BEA), and National Establishment Time Series (NETS).²⁰ The Plan must be developed based on the latest available forecast, which is required to be updated and used in the air quality conformity analysis. The following section provides a high-level overview of the forecasting process, methodologies, and results. Detailed results and descriptions of forecasting methodologies can be found in the DVRPC publication, "2050 v2.1 Population and Employment Forecasts, 2020–2050 Analytical Data Report. (Publication #26105).

The forecasts are developed in collaboration with county planning partners through the Socioeconomic and Land Use Analytics Committee (SLUAC): a group of agency staff from around the region tasked with demographic and economic analysis in their roles at their respective agencies. The SLUAC convenes to discuss, review, and advise on several DVRPC initiatives. Committee members provided comments on the proposed methodology and formed a consensus around the final forecasts. Planning partners are involved in every stage of

forecast development through the DVRPC facilitated Socioeconomic and Land Use Analytics Committee (SLUAC). The committee is made up of county planning staff and other planning partners with expertise in demographic, economic, and/or development analysis to provide input and give feedback on forecast methodology, assumptions, and data sources; county and municipal-level trends and insights; and final forecast numbers.

Population and employment changes are forecast in five-year increments at the municipal, traffic analysis zone (TAZ), county, and regional level. Population forecasts are further presented in five-year age-cohorts based on recent birth, death, and migration rates. Tables B-1 and B-2 summarize DVRPC's adopted regional and county-level population forecasts through 2050. Sub-county forecasts are available for viewing and download via:

- Web map:²¹ An interactive platform with municipal/district and county-level charts.
- Table Downloads:²² Access the DVRPC Data Catalog to preview and download tables and GIS features at county, municipal/district, and TAZ levels.

Population Forecast

The 2050 v2.1 Population Forecast provides an updated outlook on how Greater Philadelphia's population may grow and change over the next three decades. It accounts for long-term

²⁰ V1.0 was initially updated with v2.0, but more recent data became available during this Plan update, resulting in v2.1.

²¹ *Population and Employment Forecast Web Map*, DVRPC, accessed June 8, 2025, www.dvrpc.org/webmaps/popempforecasts/

²² *Adopted 2050 v2.0 Population and Employment Forecasts*, DVRPC, accessed June 8, 2025, catalog.dvrpc.org/dataset/adopted-2050-v2-0-population-employment-forecasts

demographic trends, including falling birth rates, an aging population, and the role of migration in sustaining growth through 2050. This version builds on earlier modeling while acknowledging current data limitations at the local level and sets the stage for future refinements in the next forecast cycle.

Population Methodology

The 2050 v2.1 population forecast incorporates updated base data from the 2020 Decennial Census and a new age-cohort model to project population at the county level. The primary methodological update was at the county and regional level.

The age-cohort model draws on historic decennial census data and birth and death records from the New Jersey and Pennsylvania state health departments. It projects future population by tracking the behavior of age-sex cohorts from a base year to a horizon year, using assumptions about birth, death, and migration rates. Starting with the 2020 Census as the base year, each county-level model generated five-year forecasts through 2050. At the end of each five-year cycle, cohorts are advanced to the next age group. New births are added as the 0–4-year-old age group, and the model is rerun for the next period, continuing until 2050.

Municipal forecasts were derived by aligning the 2050 v1.0 municipal forecasts with the updated 2050 v2.1 county forecasts. The proportion of county growth attributed to each municipality in the v1.0 forecast was applied to the total county growth in v2.1. This approach ensured that the sum of municipal forecasts matched the county-level control totals in 2050 v2.1.

The municipal forecasts in 2050 v1.0 were originally developed using UrbanSim, a land use model that incorporates demographic trends, housing costs, employment sector clustering, and location preferences of residents and businesses in response to changes in transportation access. The UrbanSim model was informed by a regional development pipeline, incorporating CoStar real estate data that was refined by county reviews of building permits and active construction projects.

UrbanSim was not available for update in this forecast cycle, and no alternative land use models were readily accessible. As a result, municipal-level forecasts from 2050 v1.0 were used to estimate future municipal population and household growth based on the updated county-level age-cohort projections. DVRPC intends to return to a land-use modeling approach in the next forecast cycle.

Population Results

The region's population is projected to grow by 7.8 percent—an increase of 458,783 people—between 2020 and 2050 (see Table B-1). This growth rate is like the 2050 v1.0 forecast, which projected a 7.6 percent increase, or 440,188 additional residents. The total forecasted population for 2050 in version 2.0 is higher (6,351,893) than in version 1.0 (6,206,332), largely due to the higher starting point established by the 2020 Census.

Regional population growth is expected to slow after 2035, declining from an average annual growth rate of 0.4 percent to just 0.1 percent. This deceleration is primarily driven by falling birth rates and an aging population. Despite these trends, net

migration is projected to remain high enough to offset population decline and sustain modest growth through 2050.

The 2020 Census led to adjustments in assumptions about net migration made in the v1.0 model. The census counted nearly 150,000 more residents than the 2019 Census Population Estimates had projected, suggesting that net migration during the 2010s was stronger than previously understood. Additionally, the SLUAC noted increased approvals of multifamily housing developments and higher international migration in the early 2020s as contributing factors to recent population growth.

The region's aging trend is expected to continue. In 2020, about one in six Greater Philadelphia residents were aged 65 or older. By 2050, that proportion is projected to increase to nearly one in

four. Mortality rates for most age groups are expected to decline after returning to pre-pandemic levels by 2025. However, birth rates have steadily fallen below replacement levels. Regional births are projected to peak around 2030 and then decline throughout the remainder of the forecast period.

Figures B-1, B-2, and B-3 illustrate different aspects of the 2050 population forecast:

- Figure B-1: Absolute change from 2020 to 2050
- Figure B-2: Percent change from 2020 to 2050
- Figure B-3: Total forecasted population in 2050

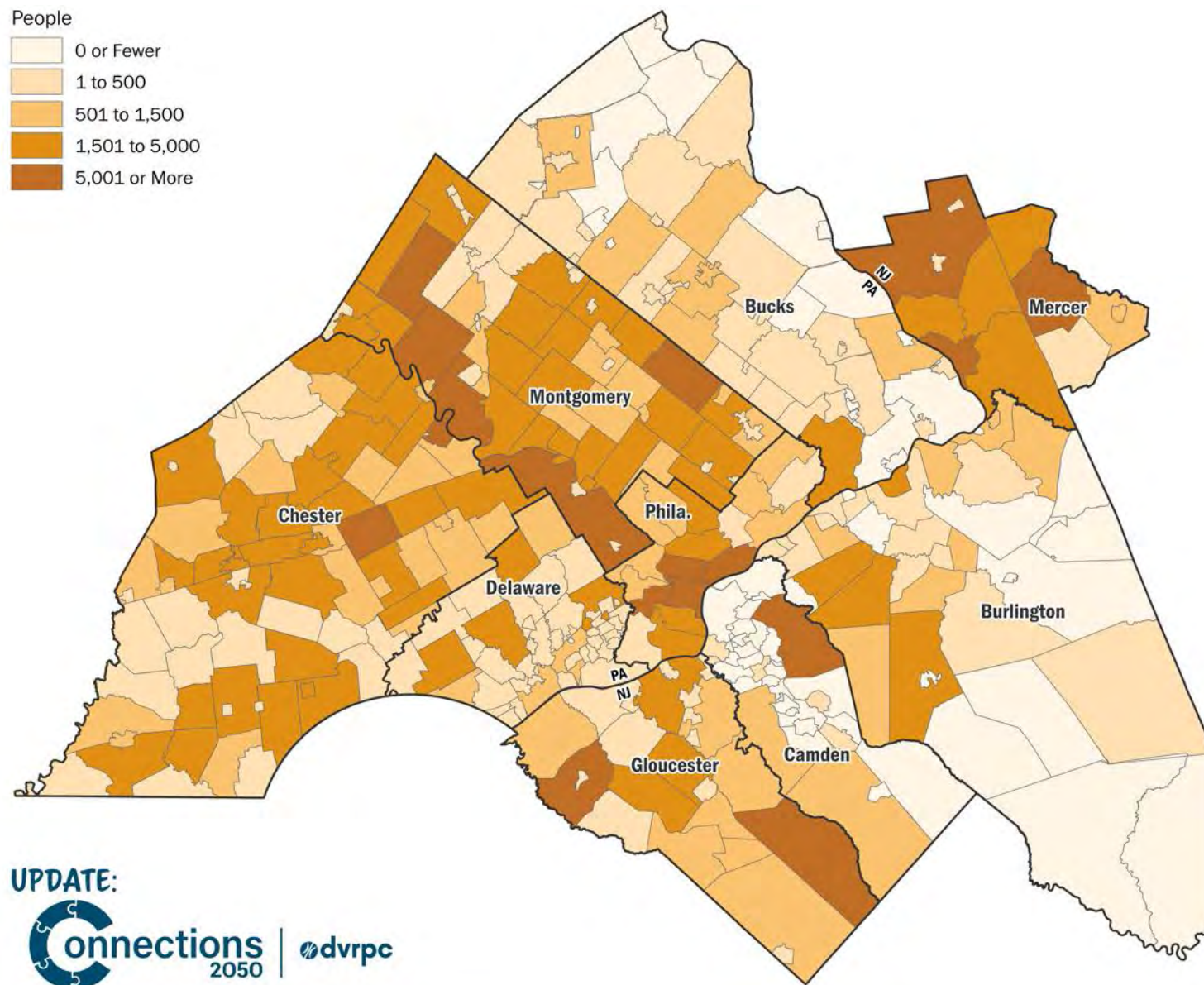
Table B-1: Forecasted Population by County (2020–2050)

County	2020	2025	2030	2035	2040	2045	2050	Absolute Change, 2020–2050	Percent Change, 2020–2050
Burlington	461,860	474,938	481,892	485,221	486,310	484,543	481,500	19,640	4.30%
Camden	523,485	529,829	531,962	534,490	535,325	532,961	529,692	6,207	1.20%
Gloucester	302,294	306,671	310,786	317,901	322,160	327,296	330,205	27,911	9.20%
Mercer	387,340	398,254	411,630	416,247	419,761	421,736	423,029	35,689	9.20%
New Jersey	1,674,979	1,709,692	1,736,270	1,753,859	1,763,556	1,766,536	1,764,426	89,447	5.30%
Bucks	646,538	653,800	660,122	664,092	664,508	661,478	655,736	9,198	1.40%
Chester	534,413	564,292	585,266	601,696	615,751	626,434	634,012	99,599	18.60%
Delaware	576,830	584,199	591,408	596,379	598,727	598,641	597,100	20,270	3.50%
Montgomery	856,553	881,522	905,095	926,337	943,123	955,916	965,342	108,789	12.70%
Philadelphia	1,603,797	1,649,774	1,681,971	1,699,155	1,711,201	1,720,856	1,735,278	131,481	8.20%
Pennsylvania	4,218,131	4,333,587	4,423,862	4,487,659	4,533,310	4,563,325	4,587,468	369,337	8.80%
DVRPC Region	5,893,110	6,043,279	6,160,132	6,241,518	6,296,866	6,329,861	6,351,894	458,784	7.80%

Source: DVRPC 2050 v2.1 Population Forecasts, 2025.

Base populations from the U.S. Census Bureau Population Estimates Program (2023 release).

Figure B-1: Absolute Population Change (2020–2050)



Source: DVRPC, 2025.
Base populations from
U.S. Census Bureau
Population Estimates
Program (2023 release).

Figure B-2: Percent Population Change (2020–2050)

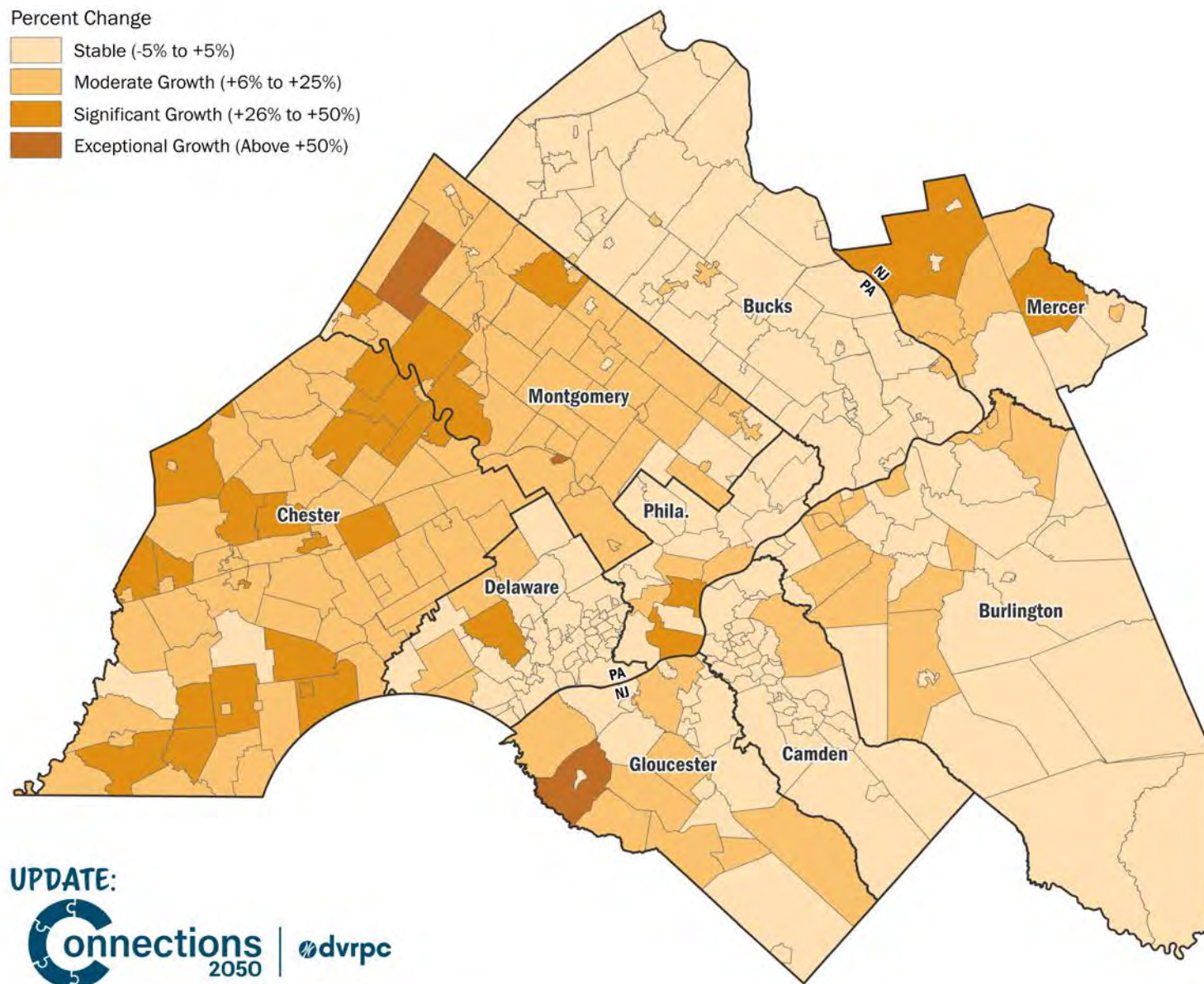
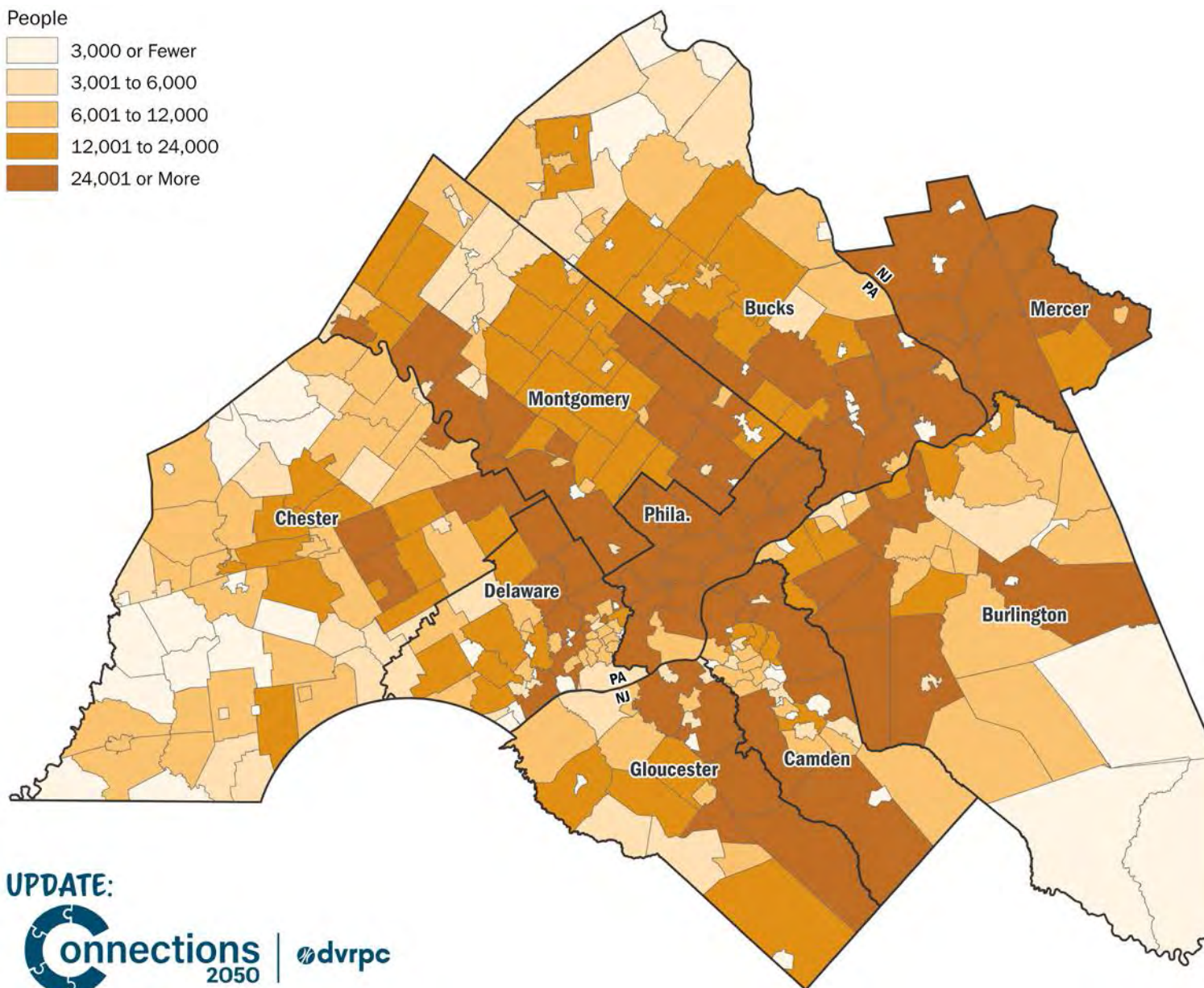


Figure B-3: 2050 Municipal Population Forecast



Source: DVRPC, 2025.
Base populations from
U.S. Census Bureau
Population Estimates
Program (2023 release).

Employment Forecast

The 2050 v2.1 Employment Forecast presents an updated outlook for job growth across Greater Philadelphia, using revised base data from the U.S. Bureau of Economic Analysis (BEA), and National Establishment Time Series (NETS) and an updated county-level model.²³ Like the population forecast, it builds on prior land use modeling to estimate employment change at the municipal and regional scales. Forecasts reflect post-COVID recovery trends, shifts in sectoral demand, and long-term demographic changes that limit labor force growth. While employment is projected to increase by over 350,000 jobs by 2050, most of this growth occurs early in the forecast period, with steady but slower gains thereafter. The DVRPC Board adopted the 2050 v2.1 Employment Forecast to better reflect more recently projected job growth from emerging and existing job centers in Philadelphia's Bellwether District, Philadelphia International Airport, and the Navy Yard.

Employment Methodology

As with the population forecast, the employment forecast update involved developing a new county-level model and leveraging results from the 2050 v1.0 UrbanSim model for municipal forecasts.

The base year for the 2020 employment forecast used data from the U.S. Bureau of Economic Analysis (BEA), which includes full-time and part-time employment figures by North American

Industry Classification System (NAICS) industry. To establish the municipal employment base, the NETS—a point-based database of jobs and business locations—was aggregated to the municipal level. This point-level NETS data was then uniformly scaled at the county level, by industry, to align with BEA county employment totals.

The 2050 v2.1 forecasts incorporated land use model results from the v1.0 forecast cycle to estimate municipal and traffic analysis zone (TAZ) growth. In v1.0, the land use model allocated projected growth at the census block level using both stakeholder-identified development data and an agent-based simulation model. This simulation incorporated various land use and socioeconomic inputs to distribute employment growth spatially.

For the v2.1 update, county-level employment forecasts were distributed to the municipal and TAZ levels in proportion to their respective growth shares from the v1.0 land use model. This ensured consistency with prior spatial patterns of employment growth while updating the overall county-level employment trajectory.

Employment Results

Employment in the DVRPC region is forecast to increase by 356,537 jobs between 2020 and 2050, a gain of 10.1 percent (see Table B-2). This includes both full-time and part-time wage or salary workers, as well as the self-employed.

²³ Both the NETS and BEA datasets have since been discontinued.

Employment growth over the forecast period is smaller than the projected population increase of 458,784 people. This difference is largely due to the aging population. Although the region's total population is expected to grow, the working-age population remains relatively stable between 2020 and 2050, limiting the potential for employment expansion. Employment increases the most between 2020 and 2025, reflecting a strong post-COVID recovery and low unemployment rates. After 2025, regional employment levels are projected to remain relatively steady.

By 2050, Educational Services, Health Care, and Social Assistance are expected to remain the largest employment sectors in the region (see Table B-3). This sector is well-positioned for continued growth due to increasing demand for health services associated with an aging population. Together with Professional, Scientific, and Technical Services and Finance, Insurance, and Real Estate, these three sectors are projected to account for nearly half of all regional jobs in 2050.

The fastest-growing employment sector, by both percent and absolute change, is Arts, Entertainment, and Recreation. The

Accommodation and Food Services sector was hit hard by the pandemic, and more than half (55 percent) of its projected growth represents a post-COVID rebound. Professional, Scientific, and Technical Services are projected to experience the second-largest job gain—adding nearly 90,000 employees, a 15.3 percent increase. Transportation, Warehousing, and Utilities rank third in absolute growth and second in percentage growth, driven by the continued expansion of e-commerce and rising demand for logistics and supply chain services. Together, these four sectors account for 85 percent of total regional employment growth through 2050.

Figures B-4, B-5, and B-6 provide visual summaries of the 2050 employment forecast:

- Figure B-4: Absolute change from 2020 to 2050
- Figure B-5: Percent change from 2020 to 2050
- Figure B-6: Total forecasted employment in 2050

Table B-2: Municipal Population Forecast by County (2020–2050)

County	2020	2025	2030	2035	2040	2045	2050	Absolute Change, 2020–2050	Percent Change, 2020–2050
Burlington	272,364	301,478	297,093	293,741	293,384	303,632	304,810	32,446	11.90%
Camden	264,617	293,858	289,958	286,678	289,606	297,805	299,050	34,433	13.00%
Gloucester	148,182	171,517	169,419	167,718	167,858	171,601	173,114	24,932	16.80%
Mercer	285,580	308,959	303,685	300,383	305,709	308,075	309,670	24,090	8.40%
New Jersey	972,763	1,077,837	1,062,185	1,050,555	1,058,597	1,083,158	1,088,694	115,931	11.90%
Bucks	361,373	390,310	384,794	380,395	384,294	381,381	383,012	21,639	6.00%
Chester	342,950	394,676	389,954	386,132	390,172	385,782	390,188	47,238	13.80%
Delaware	312,220	338,314	333,421	329,732	332,611	336,167	337,581	25,361	8.10%
Montgomery	670,496	731,893	721,105	713,129	722,970	723,543	728,787	58,291	8.70%
Philadelphia	888,524	944,294	932,071	921,523	933,628	967,601	976,631	88,107	9.90%
Pennsylvania	2,575,563	2,799,487	2,761,345	2,730,911	2,763,675	2,794,474	2,816,199	240,636	9.30%
DVRPC Region	3,546,306	3,875,299	3,821,500	3,779,431	3,820,232	3,875,587	3,902,843	356,537	10.10%

Source: DVRPC 2050 v2.1 Employment Forecasts, 2025.

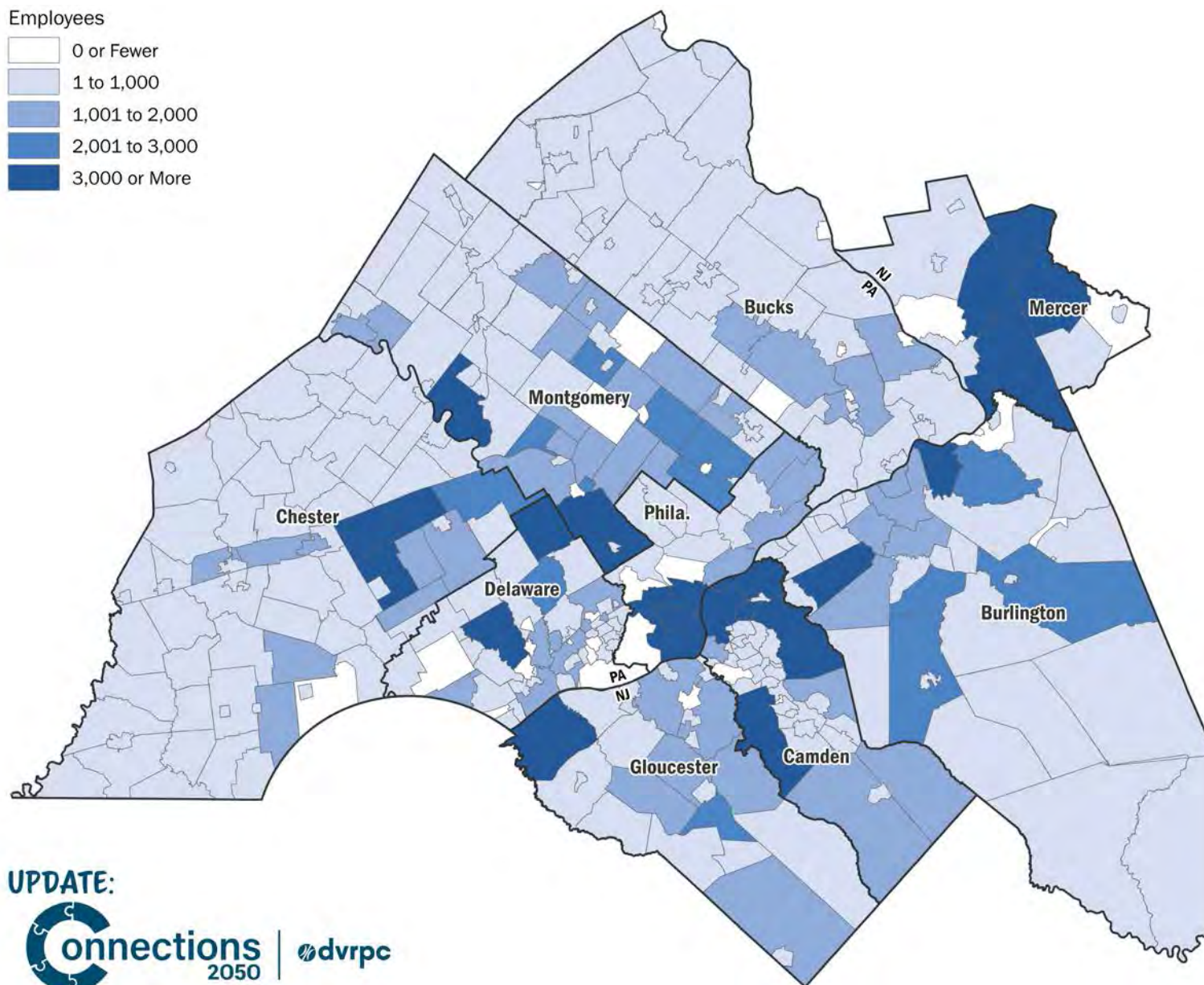
Base employment data from the National Establishments Time Series (NETS) database and Bureau of Economic Analysis (BEA).

Table B-3: Employment Forecast by Sector (2020–2050)

Employment Sector	2020	2050	Absolute Change, 2020–2050	Percent Change, 2020–2050
Agriculture and Mining	16,431	15,405	-1,026	-6.2%
Construction	160,659	186,848	26,189	16.3%
Manufacturing	176,476	163,520	-12,956	-7.3%
Wholesale Trade	119,081	105,246	-13,835	-11.6%
Retail Trade	302,164	319,774	17,610	5.8%
Transportation and Warehousing, Utilities	189,367	262,938	73,571	38.9%
Information	61,987	64,749	2,762	4.5%
Finance and Insurance, Real Estate	443,505	428,049	-15,456	-3.5%
Professional, Scientific, and Technical Services	588,473	678,320	89,847	15.3%
Educational Services, Health Care and Social Assistance	722,982	776,794	53,812	7.4%
Arts, Entertainment, and Recreation; Accommodation and Food Services	239,029	380,289	141,260	59.1%
Other Services	175,844	184,458	8,614	4.9%
Public Administration	350,297	336,456	-13,841	-4.0%
Total Employment	3,546,295	3,902,846	356,551	10.1%

Source: DVRPC, 2025.

Figure B-4: Absolute Employment Change (2020–2050)



*Source: DVRPC, 2025.
Base populations from
U.S. Census Bureau
Population Estimates
Program (2023 release).*

Figure B-5: Percent Employment Change (2020–2050)

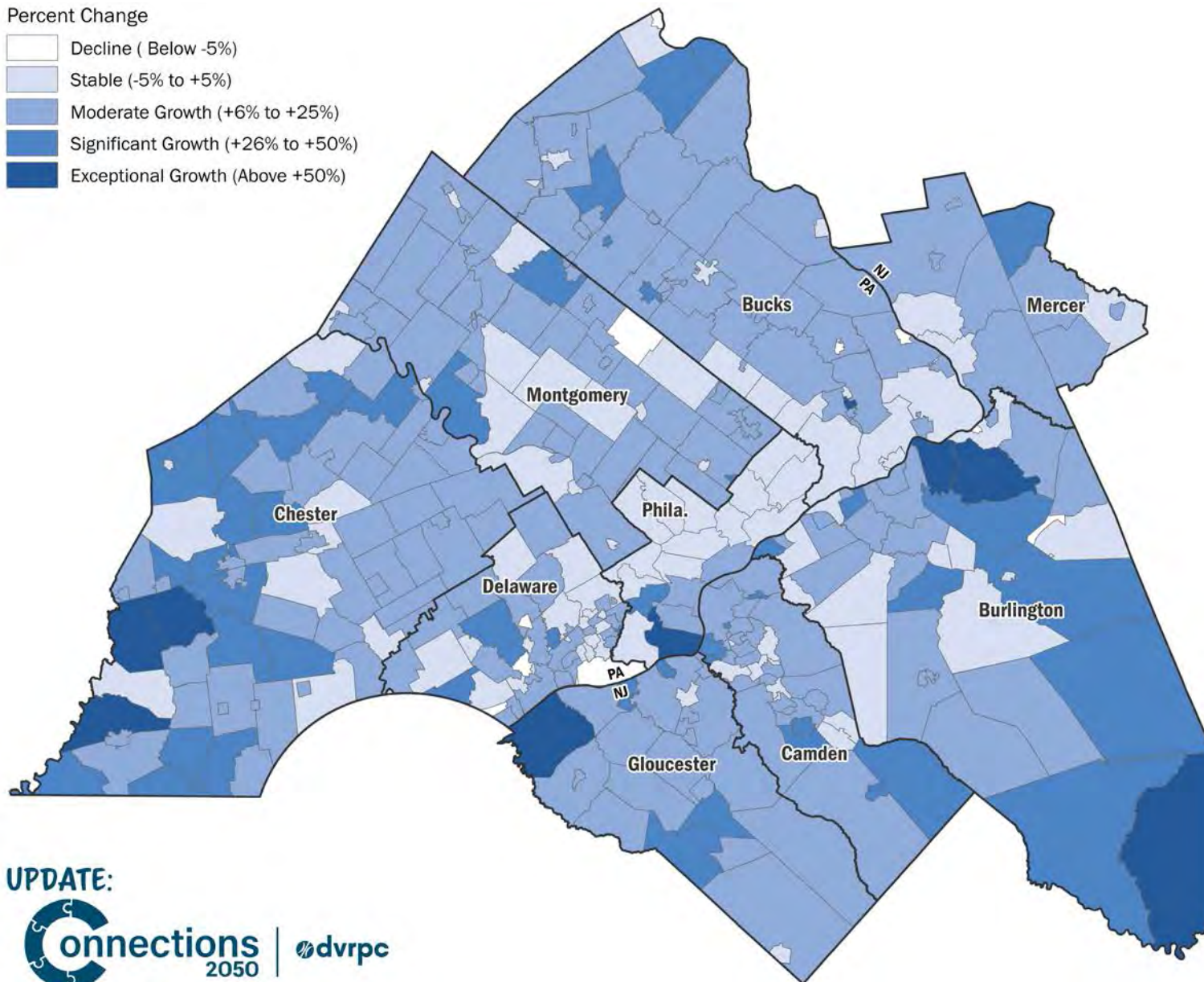
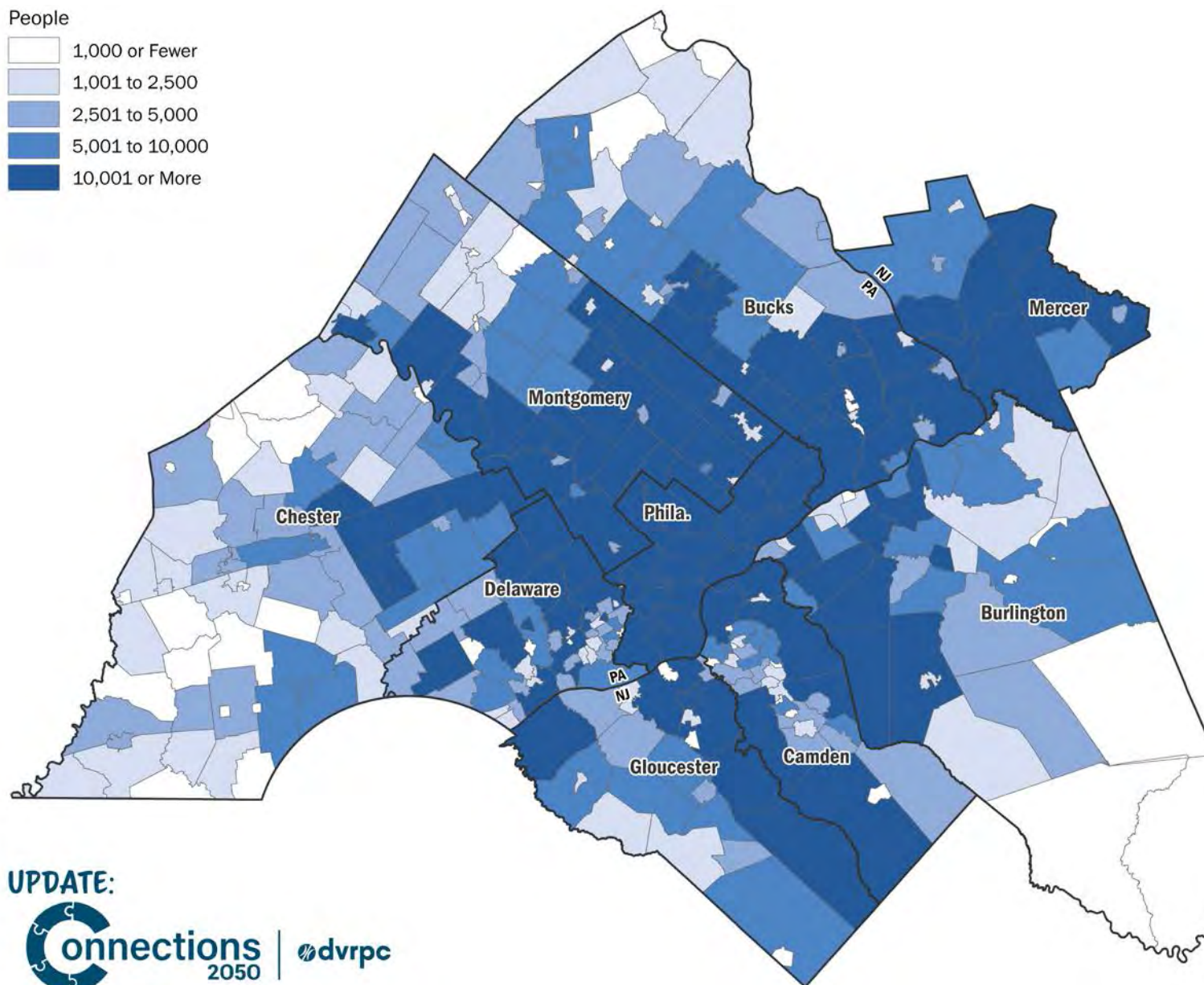


Figure B-6: 2050 Municipal Employment Forecast



Source: DVRPC, 2025.
Base populations from
U.S. Census Bureau
Population Estimates
Program (2023 release).

Title VI by Census Tract

DVRPC uses its Title VI Compliance Tool analysis across its programs to help demonstrate compliance with Title VI of the Civil Rights Act of 1964, which states:

“No person in the United States shall, on the grounds of race, color, or national origin, be excluded from participation in, be denied the benefits of, or be subjected to discrimination under any program or activity receiving federal financial assistance.”

DVRPC follows guidance and methodologies from the Federal Highway Administration (FHWA), Federal Transit Administration (FTA), Pennsylvania Department of Transportation (PennDOT), and New Jersey Department of Transportation (NJDOT) to ensure that the needs and perspectives of diverse populations are represented in planning and decision-making processes. The Title VI methodology is one component of a broader strategy that also includes public participation, stakeholder engagement, data analysis, and other research used to support collaborative planning for all residents of Greater Philadelphia.

Initially developed by DVRPC in 2001, this analysis has since evolved and been adopted or adapted by peer agencies across

the country. It is widely recognized as a best practice for demonstrating compliance with federal nondiscrimination requirements.

The Title VI analysis identifies populations protected under Title VI using data from the U.S. Census Bureau’s American Community Survey (ACS) five-year estimates. These populations are mapped at the census tract level using Geographic Information Systems (GIS) to show spatial distribution across the region (see Figure B-7). Each identified population group functions as an “indicator” in the analysis, as listed in Table B-4. Data for each census tract in Figure B-7 is the sum of the concentration of each of the nine population groups in Table B-4 based on a five-point scale ranging from well above average (4 points), above average (3 points), average (2 points), below average (1 point), and well below average (0 points). Well above average is determined based on being two standard deviations above the regional mean, above average by being one standard deviation from regional mean, average being within one standard deviation from the regional mean, below average being less than one standard deviation from the regional mean, and well below average being more than two standard deviations below the regional mean.

Figure B-7: Title VI Populations by Census Tract

Communities of Concern Concentrations
(All Indicators Combined)

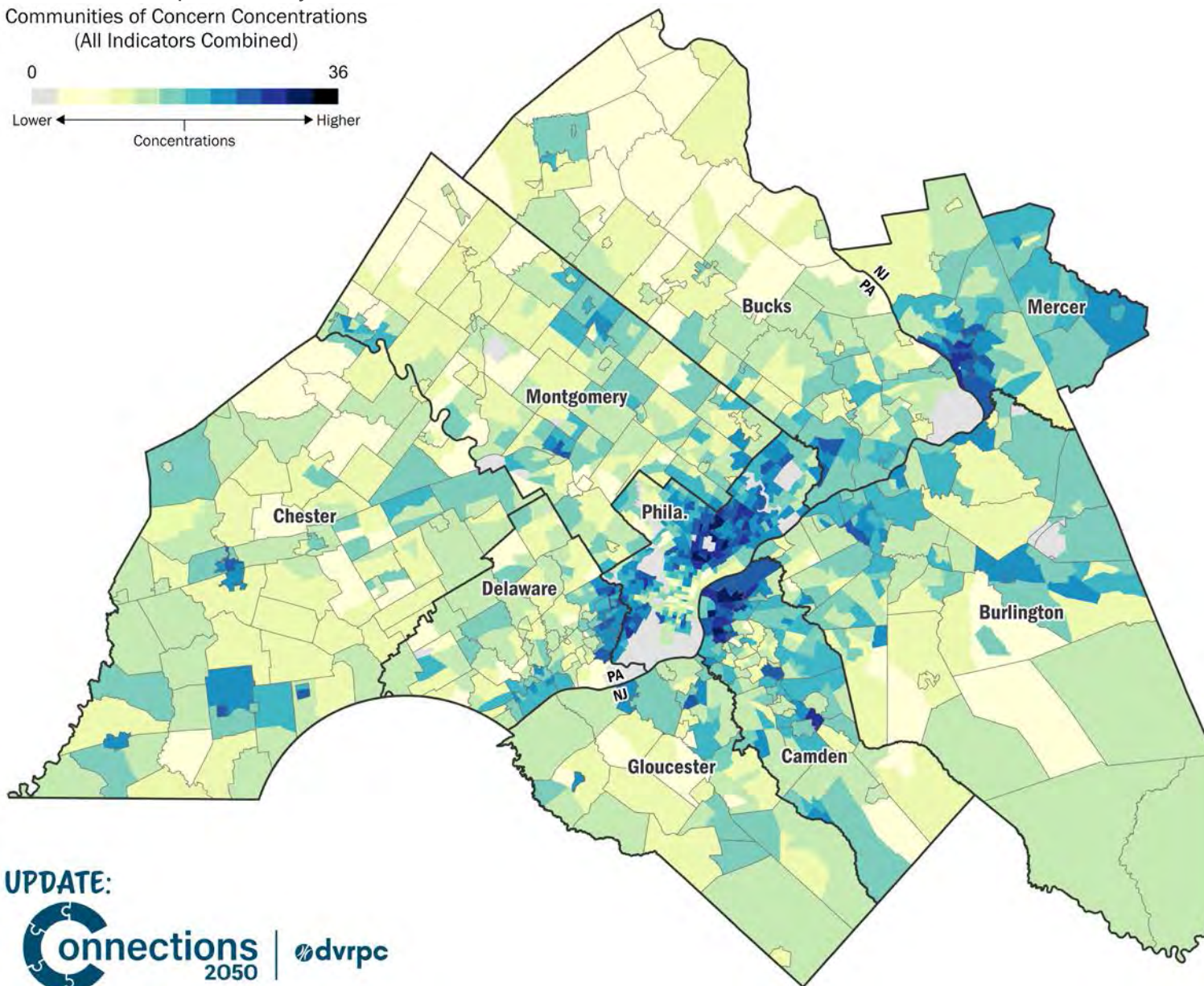
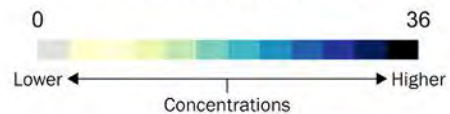


Table B-4: Title VI Population Groups and Data Sources

Indicator	ACS Data Table	Protected Population	Authorizing Source
Ethnic Minority	B03002: Hispanic or Latino Origin by Race	Minority and National Origin	Title VI of the Civil Rights Act of 1964, FHWA Title VI, and Title VI Requirements and Guidelines
Female	S0101: Age and Sex	Sex	FHWA Title VI
Foreign Born	B05012: Nativity in the United States	National Origin	Title VI of the Civil Rights Act of 1964, FHWA Title VI, and Title VI Requirements and Guidelines
Limited English Proficiency	S1601: Language Spoken at Home	Limited English Proficiency and National Origin	Title VI of the Civil Rights Act of 1964, FHWA Title VI, and Title VI Requirements and Guidelines
Low-Income	S1701: Poverty Status in the Past 12 Months	Low-Income	FHWA Title VI
Older Adults	S0101: Age and Sex	Age	FHWA Title VI
People of Color	B02001: Race	Race and Minority	Title VI of the Civil Rights Act of 1964, FHWA Title VI, and Title VI Requirements and Guidelines
People with Disabilities	S1810: Disability Characteristics	Disability	FHWA Title VI
Youth	B09001: Population under 18 Years by Age	Age	FHWA's Title VI Program and Additional Nondiscrimination Requirements (FHWA Title VI)
Ethnic Minority	B03002: Hispanic or Latino Origin by Race	Minority and National Origin	Title VI of the Civil Rights Act of 1964, FHWA Title VI, and Title VI Requirements and Guidelines
Female	S0101: Age and Sex	Sex	FHWA Title VI

Source: DVRPC, 2025.

Development and Housing

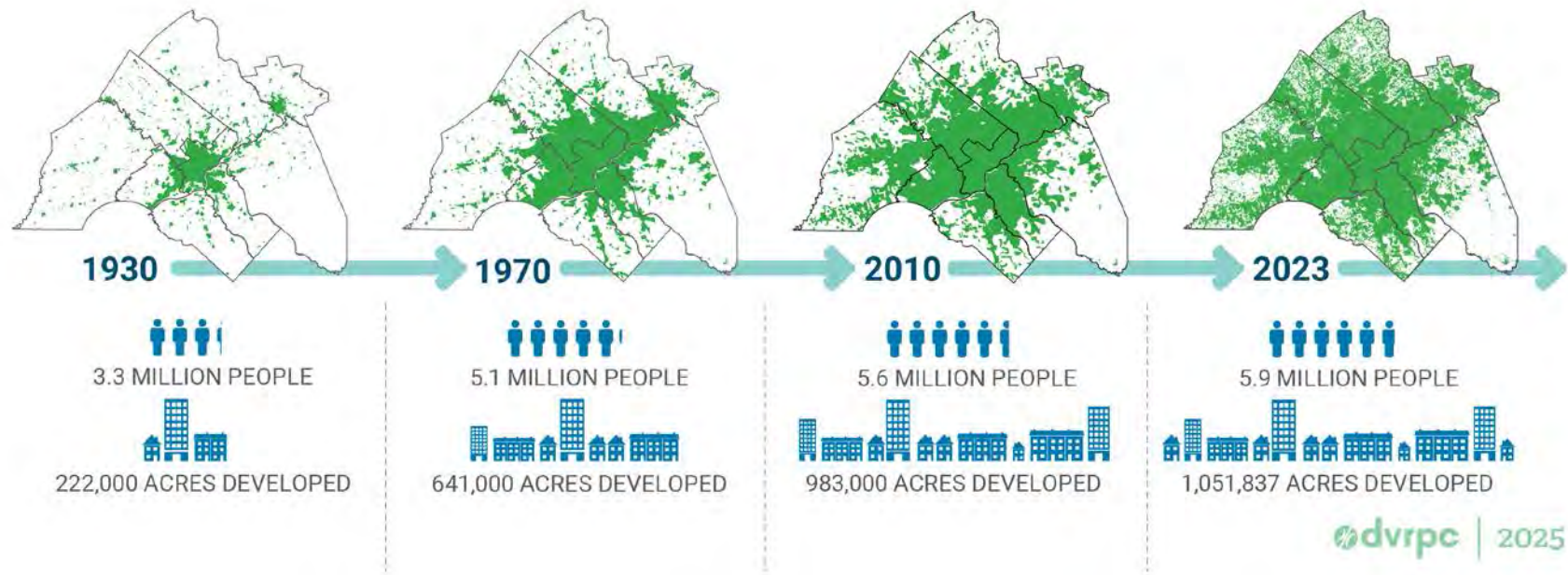
Technological advancement, economic shifts, and demographic changes have all played a significant role in shaping the character and community form of Greater Philadelphia. Over the 20th century, the region evolved from an industrial economy dominated by urban areas to one characterized by suburban expansion and increased reliance on automobiles. For example, in 1900, almost two-thirds of the region's population lived within the City of Philadelphia. By 2000, Philadelphia accounted for just 28 percent of the region's population. This decentralization of population and employment has profound implications for how DVRPC and its planning partners coordinate land use and transportation planning.

The maps in Figure B-8 help to illustrate how land use and development have evolved in the region since 1930. In 1930, development within the region was largely centered on Philadelphia's urban core. At that time, approximately 3.3 million people lived in the nine counties, and roughly 222,000 acres in the region were developed. By 1970, the region's population increased to 5.1 million people, and the amount of developed land nearly tripled to 641,000 acres. Over those four decades, the rate at which land was developed grew three and a half times faster than the rate of population increase.

This trend accelerated between 1970 and 2010, when land consumption increased at roughly five and a half times the rate of population increase. These shifting development patterns have increased our resource needs, strained our transportation systems, and contributed to economic and racial segregation—challenges that our region is still grappling with.

The rate at which land development has consumed new land has slowed significantly in recent years. Between 1990 and 2010, an average of 9,050 acres of land were developed annually regionwide. Between 2010 and 2023, this figure dropped to roughly 3,600 acres of new development per year. There has also been a significant shift in the location of new development from the turn of the 21st century to the present day. Since 1999, the region has witnessed a 46 percent increase in the number of residential permits approved in areas characterized as Core Cities and Developed Communities. Focusing new growth in areas such as these, with existing transportation and utility infrastructure, remains a major focus of the Long-Range Plan. At the regional level, smart growth works by directing new development toward existing communities where it can help to leverage existing investments while conserving valuable open space and natural resources.

Figure B-8: Extent of Regional Development (1930–2023)



Source: Base populations from the U.S. Census Bureau Population Estimates Program (2023 release).

Note: The data collection process was revised between 2010 and 2015. As a result, figures from before and after these years may not be directly comparable.

Housing

Housing affordability has become an increasingly pressing issue for the residents of Greater Philadelphia in recent years, just as it has for Americans as a whole. Housing costs have been rising faster than incomes, and the demand for lower-cost housing has far outstripped the supply. According to the Joint Center for Housing Studies of Harvard University, the median sales price for existing single-family homes was 4.9 times the median household income in 2023. This figure represents a significant

jump from the 4.1 price-to-income ratio of 2019 and the 3.2 ratio averaged during the 1990s.

What had been a long-term concern grew into a full-blown crisis during the COVID-19 pandemic as home prices surged throughout much of Greater Philadelphia. Between February 2020 and February 2025, the median sale price of homes in Burlington, Camden, Gloucester, and Mercer counties all increased by over 80 percent, according to sales data from Redfin. Median sale prices increased by over 40 percent in

Bucks, Chester, Delaware, and Montgomery counties during the same period. It is no surprise, then, that 755,000 households (one in three households/33 percent of all households) in the region spent 30 percent or more of their income on housing. This threshold officially classifies them as “cost burdened.” The problem is particularly acute for residents who rent their homes—over 52 percent of renter households are cost-burdened.

As the cost of housing squeezes household budgets, many families are being forced into precarious living situations that affect their health, the length of their commute, and their ability to save for emergencies or make long-term investments. Furthermore, rising home prices, elevated interest rates, and a limited supply of homes for sale have pushed homeownership out of reach for many, eliminating or delaying opportunities to build generational wealth and reducing access to economic opportunity.

The region’s affordability challenges are complex and multifaceted; yet many of our problems can be traced to the long-term underproduction of housing and the resulting mounting shortage. Analysis conducted by Up for Growth (UFG), a research and advocacy organization sponsored by the American Planning Association, suggests that the 11-county Philadelphia Metropolitan Statistical Area was missing nearly 80,000 housing units in 2021, the 10th highest total of any metropolitan area in the country. Analysts cite pandemic-influenced economic conditions, labor shortages, financing availability, and regulatory barriers to new development as key contributors to the shortage. Over time, these supply constraints

have contributed to increasing housing costs and reducing affordability.

Taking a regional approach to solving our housing challenges is essential because the conditions that influence the cost and supply of housing do not neatly align with municipal or county boundaries, and no one community can solve it alone. Ensuring that there is an adequate supply of quality, affordable, and accessible homes in our region will require expanding housing production, mitigating displacement, preserving existing housing, and promoting collaboration between municipalities. In 2022, DVRPC launched its Regional Housing Initiative to investigate the nature and extent of the housing challenges facing communities in our region and how we can work together to address them. As part of this initiative, DVRPC staff conducted research, gathered data, and engaged with stakeholders and subject matter experts, including county and state planners, for-profit and nonprofit developers, and housing advocates across a wide variety of topics. For more information on this effort, visit

www.dvrpc.org/housing/regionalhousinginitiative.

Although many factors contributing to the housing crisis are beyond the control of local governments, municipalities do have control over where, what type, and how much housing can be built within their borders. In many places, zoning and other land use regulations have systematically prioritized low-density single-family detached homes, making it difficult or impossible to build in-demand, lower-cost housing types, such as small lot single-family homes, missing middle housing, and multifamily units. Accordingly, zoning reform aimed at expanding housing

choice and supply is one of the most critical ways that local governments can directly combat the affordability crisis.

While increasing the supply and range of housing options is crucial, where we build is just as important as what we build. Focusing new development in high-opportunity neighborhoods—places rich in jobs, infrastructure, transportation, and community assets—can help our region address other critical regional issues, including transportation, economic development, environmental protection, and resilience. Prioritizing locations near existing, high-quality transit further supports smart growth by reducing car dependence, expanding access to opportunity, and making more efficient use of public infrastructure. *Update: Connections 2050* revises the Plan’s Centers to better align them with existing density and high-quality transit service.

Centers

Update: Connections 2050 places Centers at the heart of its vision for the region’s future, using them as a foundational framework to guide regional growth, development, and transportation investment in smart, efficient ways to make the most of existing assets. By linking land use and transportation planning, Plan Centers help direct resources to where they can have the greatest impact, cost-effectively. These are locations of local government, essential services, economic activity, and community life. In addition to Plan Centers, the Plan identifies Freight Centers—strategic areas that reflect and support the region’s freight-based land use and economic development patterns.

Plan Centers

Centers have been a foundational element of DVRPC’s long-range planning efforts since the adoption of the *Direction 2020* Plan in 1992, and the concept appeared in even earlier plans. Centers are the organizing principle of *Update: Connections 2050*—focal points in the regional landscape that shape how and where growth occurs. They provide a structure for coordinating land use with infrastructure investments, enabling the most efficient provision of transportation, water, sewer, and other critical services. Prioritizing growth in and around Centers supports the Plan’s goals by preserving open space, protecting natural resources, and fostering vibrant, walkable, and connected communities. By emphasizing Centers, the Plan promotes an accessible and more efficient regional development pattern that enhances quality of life for all.

Plan Centers are places in the region that are, or do, one or more of the following:

1. Highlight and showcase what good development looks like in the regional context.
2. Support higher levels of density for future growth through mixed-use, infill development.
3. Serve as priority areas for investment, particularly by building and enhancing multimodal transportation infrastructure. This, combined with mixed-use, high-density development patterns, can shorten trip lengths and improve access through transit, walking, and biking.

Centers Update

As part of the *Update: Connections 2050* Plan, DVRPC staff conducted background research to support a major revision of the region's designated Centers. This effort included a review of how Plan Centers have been incorporated in previous long-range plans, land use planning tools used by peer agencies, and recent national research on activity centers. Based on this analysis, DVRPC reaffirmed its commitment to keeping Centers as a core element of the regional plan and identified strategies to modernize and strengthen the Centers framework going forward.

The update to the Centers framework focused on three key goals:

1. **Clarify the role of Centers** and how they support the implementation of the regional plan.
2. **Revise Center types** to provide more specific recommendations tailored to each category.
3. **Incorporate more quantitative data** into the definition and analysis of Centers by aligning Center boundaries with U.S. Census geographies, such as clusters of block groups, municipalities, or census-designated places.

The third goal will allow DVRPC to conduct more detailed research and develop targeted policy recommendations for each Center. Over the longer term, it will also support the integration of Centers into DVRPC's Tracking Progress initiative, enabling the measurement of Long-range Plan goals within these areas.

Centers Identification

The core variables that define Centers are transit access and development density. Figure B-9 illustrates how each Center category is positioned on a 2x2 matrix formed by these two variables—population and employment density on the x-axis and frequency/reliability of transit service on the y-axis. Each Center type is associated with specific quantitative thresholds, as shown in the figure.

Centers located in the top-right quadrant—with both high density and strong transit access—represent the ideal development pattern for the region and are considered existing Transit-Oriented Development (TOD) showcases. High-density areas without strong transit access are opportunities for multimodal transportation investment. Transit-rich areas lacking density are ideal candidates for TOD investment, or TOD Opportunity Centers. With the right investments—outlined in the policy priorities that follow—Centers in the latter two quadrants can shift toward becoming TOD showcases.

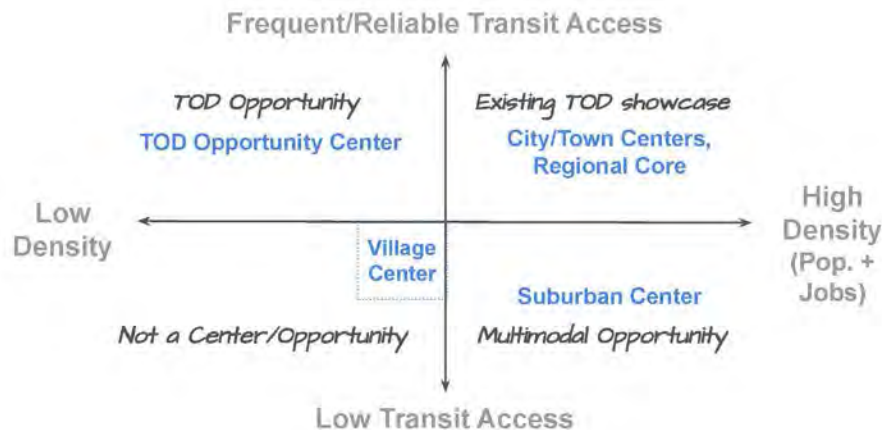
Low-density areas without transit access are typically not suited for Center designation, except for Village Centers. Village Centers serve as important and recognizable activity nodes, often within the region's rural areas. While they may not meet the density or transit thresholds found in more urbanized areas, their local significance justifies their inclusion.

Centers Categories

DVRPC collaborated with county planning partners to refine the list of Centers identified through the quantitative analysis.

Update: Connections 2050 designates 176 Centers, categorized into one of six Center types (see Figure B-10; Tables B-5 and B-6). The review analyzed municipal geographies, Census Designated Places, and census block groups. Inclusion of the local government's incorporation designation (such as city, borough, or township) in tables B-5 and B-6 indicates it covers the entire municipal geography. The update added 75 new Centers and removed 10 existing Centers that no longer met the established criteria.

Figure B-9: Centers Typology Matrix



Source: DVRPC, 2025.

The six Center categories are defined as follows:

1. **Regional Core**—The central business district of the region, encompassing Greater Center City Philadelphia, University City, and downtown Camden. Geography is determined by the highest level of existing density.

2. **City Centers**—Dense, mixed-use neighborhoods or urban core cities with strong transit access and walkable main streets. These serve as models for urban development.
3. **Town Centers**—Compact, mixed-use boroughs or census-designated places with existing or historical transit service and a walkable main street. These Centers represent ideal development in a regional (non-urban) context.
4. **Suburban Centers**—Areas with higher density than their surroundings or clusters of jobs and commercial activity, but limited or no direct transit access. These Centers often face a jobs-housing imbalance and are key locations for multimodal and quality-of-life investments.
5. **TOD Opportunity Centers**—Areas with access to regional rail or strong transit service that could evolve into Town Centers through Transit-Oriented Development (TOD) strategies. These strategies may include increasing density, improving walkability, and enhancing multimodal connections.
6. **Village Centers**—Moderate-density, mixed-use clusters typically located in rural or agricultural settings. They often feature an identifiable downtown or main street and are important nodes of local cultural and economic activity.

Figure B-10: Plan Centers

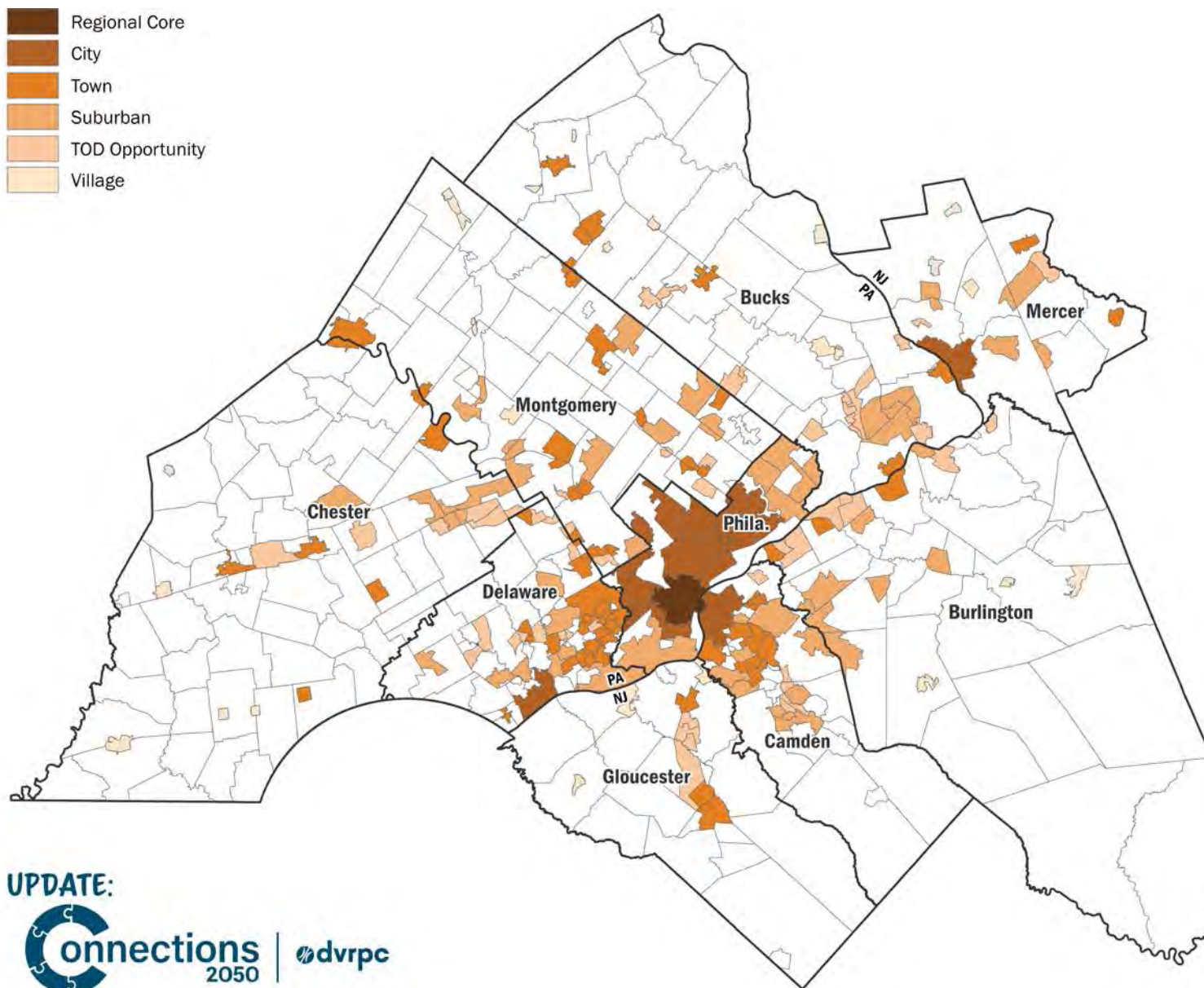


Table B-5: Pennsylvania Plan Centers

County	City Centers	Town Centers	Suburban Centers	TOD Opportunity Centers	Village Centers
Bucks		• Bristol Borough • Doylestown Borough • Morrisville Borough • Perkasie / Sellersville • Quakertown Borough • Telford / Souderton	• Fairless Hills • Levittown	• Chalfont Borough • Langhorne • Neshaminy / Trevoise • New Britain Borough • Tullytown Borough • Warminster • Woodbourne • Yardley Borough	• Dublin Borough • New Hope Borough • Newtown • Richlandtown Borough • Silverdale Borough • Trumbauersville Borough
Chester		• Coatesville • Downingtown Borough • Kennett Square Borough • Phoenixville Borough • Pottstown • Spring City / Royersford • West Chester Borough	• Great Valley • King Of Prussia • Paoli • Uwchlan	• Berwyn-Strafford Main Line • Exton • Malvern Borough • Thorndale	• Avondale Borough • Honey Brook Borough • Oxford Borough • Parkesburg Borough • West Grove Borough

County	City Centers	Town Centers	Suburban Centers	TOD Opportunity Centers	Village Centers
Delaware	Chester City	- Ardmore - Bryn Mawr - Clifton Heights Borough - Collingdale Borough - Darby Borough - Folcroft Borough - Glenolden Borough - Havertown - Lansdowne Borough - Linwood / Marcus Hook - Media - Prospect Park / Norwood - Ridley Park Borough - Sharon Hill Borough - Swarthmore Borough - Upper Darby Township - Wayne	- Airport / Lower South - Brookhaven Borough - Broomall - Concordville - Folsom - Springfield - Village Green-Green Ridge - Woodlyn	- Aldan - Elwyn - Haverford - Morton - Rosemont - Saint Davids - Villanova / Radnor - Wallingford - Wawa	

County	City Centers	Town Centers	Suburban Centers	TOD Opportunity Centers	Village Centers
		• Yeadon Borough			
Montgomery	• Fox Chase / Burholme	• Ambler Borough • Ardmore • Bryn Mawr • Conshohocken • Glenside • Hatboro Borough • Jenkintown Borough • Lansdale Borough • Narberth Borough • Norristown / Bridgeport • Pottstown • Spring City / Royersford • Telford / Souderton	• Bala Cynwyd • Collegeville • Fort Washington • Horsham • King Of Prussia • Montgomeryville • Plymouth Meeting • Willow Grove	• Elkins Park • Gulph Mills • Orelan • Rosemont • Rydal / Noble • Villanova / Radnor • Wynnewood	• Eagleville • Schwenksville Borough • Tri-Borough

County	City Centers	Town Centers	Suburban Centers	TOD Opportunity Centers	Village Centers
Philadelphia	• City Avenue / Overbrook • Fox Chase / Burholme • Frankford • Germantown / Mount Airy / Chestnut Hill • Kingsessing • Lawncrest / Oxford Circle • Manayunk / Roxborough • Mayfair / Holmesburg • North Philadelphia • Rhawnhurst / Caster Gardens • River Wards • South Philadelphia • Southwest Philadelphia • Upper North Philadelphia • West Philadelphia		• Airport / Lower South • Bustleton • Morrell Park / Millbrook • Parkwood • Somerton • Torresdale		

Source: DVRPC, 2025.

Table B-6: New Jersey Plan Centers

County	City Centers	Town Centers	Suburban Centers	TOD Opportunity Centers	Village Centers
Burlington		• Burlington City • Borough of Palmyra • Riverside Township	• Florence • Larchmont • Moorestown • Mount Holly Township • Mount Laurel / Marlton • Willingboro	• Beverly / Delanco • Bordentown City • Cinnaminson / Riverton • Roebling	• Browns Mills • Medford Lakes Borough • Pemberton Borough
Camden	• Camden City	• Barrington Borough • Collingswood Borough • Gloucester City • Haddon Heights Borough • Haddonfield Borough • Merchantville • Westmont	• Audubon Borough • Bellmawr Borough • Cherry Hill • Echelon • Lindenwold • Mount Laurel / Marlton • Runnemede Borough • Stratford Borough	• Lindenwold Station • Pennsauken • Woodcrest / Ashland	

County	City Centers	Town Centers	Suburban Centers	TOD Opportunity Centers	Village Centers
Gloucester		• Glassboro • Pitman Borough • Woodbury City		• Sewell • Woodbury Heights / Oak Valley / Wenonah	• Paulsboro Borough • Swedesboro Borough • Westville Borough
Mercer	• Trenton City	• Hightstown Borough • Princeton	• Hopewell • Mercerville • Robbinsville • Route 1 Corridor	• Ewing / West Trenton • Princeton Junction	• Hopewell Borough • Lawrenceville • Pennington Borough

Source: DVRPC, 2025.

Policy Strategies

Centers represent priority locations for infill and redevelopment, supporting a mix of uses and increased density. Higher density enables walkable communities and promotes multimodal transportation infrastructure, both of which are priorities for all Center types. Tailored policy strategies for each Center category are outlined below:

Town Centers and City Centers

These Centers serve as regional models for successful development and are positioned to advance:

- Higher-density, mixed-use, and infill development

- Pedestrian and bicycle safety through supportive infrastructure
- Historic preservation and context-sensitive development
- Strategic investment in transit and multimodal transportation integration

Suburban Centers

To evolve toward the functionality and character of Town Centers, Suburban Centers:

- Promote a better job-housing balance through:
 - Suburban retrofits

- Infill ordinances
- Neighborhood design codes or form-based zoning
- Mixed-use development and live-work spaces
- Develop a walkable main street by:
 - Investing in Complete Streets infrastructure
 - Implementing traffic calming, road diets, and increased intersection density
 - Updating zoning to allow higher-density, mixed-use, and multifamily housing

TOD Opportunity Centers

To support TOD and shift toward Town Center characteristics, strategies for these areas:

- Encourage mixed-use development and increased density through:
 - Zoning reforms to allow a mix of transit-supportive land uses and walkable design
 - Designating commercial main streets and permitting multifamily or “missing middle” housing
 - Adopting Neighborhood Design Codes, TOD zoning overlays, or form-based codes

- Creating density bonus incentives tied to affordable housing or public realm improvements
- Develop a walkable commercial main street by:
 - Investing in Complete Streets and other pedestrian infrastructure
 - Implementing traffic calming, road diets, and increasing intersection density
- Prioritize transit and multimodal integration to maximize the benefit of existing infrastructure

Village Centers

While not intended for high-density or rail transit development, Village Centers play a critical role in rural areas by concentrating services, jobs, retail, and housing. Strategies to enhance that role:

- Support mixed-use and higher-density development to reduce rural sprawl through:
 - Missing middle-housing
 - Neighborhood Design Codes or form-based zoning
 - Historic preservation
 - Live-work opportunities
 - Small-scale/artisanal manufacturing
 - Transfer of Development Rights (TDR) programs

Freight Centers

DVRPC also fully updated its freight centers in 2025 using newer data (see Figure B-11). Freight Centers are areas with a high

concentration of freight-intensive establishments, freight-intensive employment, freight-related land uses, industrial development, and intermodal freight connectors. They are classified into five specific typologies:

1. **International Gateway**—are core nodes in the regional and national goods movement system and serve as a connection to global markets. They are focused on a single or multiple air, maritime, or port facilities.
2. **Heavy Industrial**—are nodes focused on heavy industrial land uses involved in the manufacturing of goods. They are served by freight rail access and often have additional access to a port terminal, allowing for the movement of bulk or break-bulk source materials.
3. **Distribution & Logistics**—are nodes with a high concentration of regional and national serving distribution and logistics businesses. They are often located around key highway interchanges with access to both port gateways and consumer markets.
4. **High-Tech Manufacturing**—is focused on advanced manufacturing land uses and businesses such as pharmaceutical, biotechnology, or aviation businesses. They rely less on major freight rail and maritime facilities but are well located relative to highway facilities.
5. **Local Manufacturing & Distribution**—are focused on locally serving small manufacturing and distribution facilities. They are less dependent on prime location near Interstate interchanges but are well served by smaller highway facilities and proximity to consumer populations.

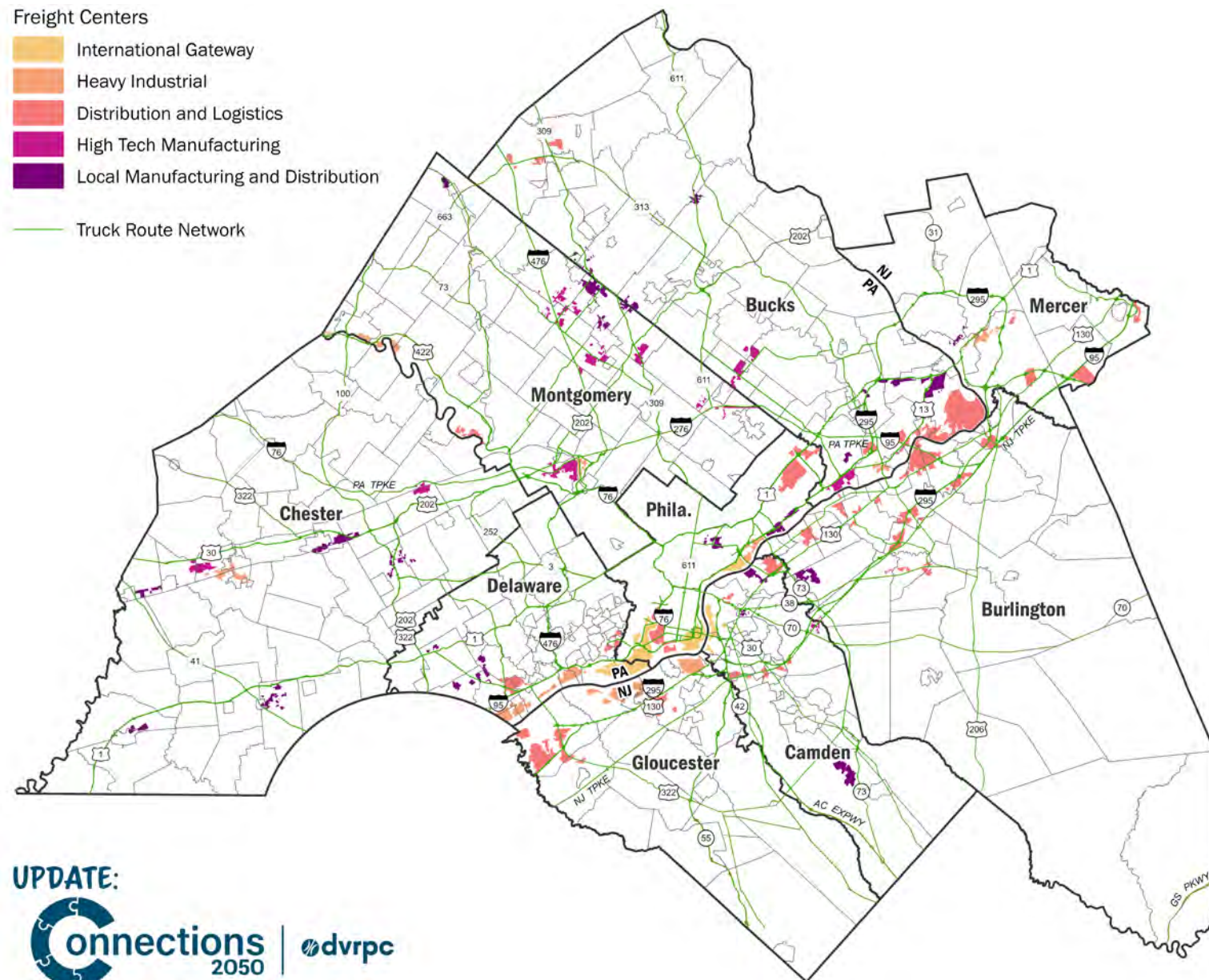
Freight Center typologies help planners, decision-makers, and other users to better understand transportation infrastructure requirements, land use and land development patterns, building types, employment characteristics, and potential community impacts that may exist at each freight center type.

Freight-Intensive Truck Network

DVRPC's Freight Program analyzed regional truck movements and Freight Centers to create a network of freight-intensive truck corridors, a key part of the upcoming Regional Freight Plan. This network will help planners address local quality-of-life concerns related to freight impacts and clearly illustrate the relationships between economic development, land use, and truck routing. Building upon an existing local framework, the analysis combined data-driven methods with essential input from county and city planners on local context and restrictions.

As part of this effort, DVRPC promotes "Freight-as-a-Good Neighbor"—an initiative that encourages cooperation across private, public, and nonprofit partners to mitigate negative externalities associated with freight. The DVRPC Data Center hosts Good Neighbor Freight features that showcase examples from across the region.

Figure B-11: Freight Centers



Source: DVRPC, 2025.

Environmental Resources

Open space, farmland, soil, water, and natural resources are indispensable to our region and its residents. Between 1930 and 2020, the population in Greater Philadelphia increased by 44 percent, while the amount of land consumed for development increased by 470 percent, resulting in significant losses of farms, fields, forests, and natural areas. This sprawling development pattern negatively impacts the environment, the economy, the transportation network, and our region's character and quality of life. The loss of healthy forested headwaters, riparian buffers, and naturally functioning floodplains, for example, degrades water quality, fragments natural habitats, decreases biodiversity, and makes our communities more susceptible to the impacts of extreme weather.

Excessive land consumption has costly consequences for local communities: increased flooding; higher costs for clean drinking water; decreases in soil productivity and nutrient cycling; reduced property values; and the loss of local agricultural economies and products. Unmanaged growth and the loss of open space also strain the region's transportation infrastructure, diminish community character, and limit opportunities for personal interaction with nature and green spaces.

Update: Connections 2050 recognizes that the loss of these resources does not offer a stable future for humans. While land consumption trends have slowed in recent decades, strengthening this trend urgently requires both growth management and open space preservation techniques. Strategic land preservation, market-based conservation, smart

growth, and enhanced community design are all needed to slow and stabilize unmanaged growth patterns at the regional scale. The Plan proposes to permanently protect at least one million of the region's 2.4 million acres of land by 2040. Preserved lands should be located in the region's Greenspace Network and Rural Resource Lands included in the Land Use Vision.

Land Use Vision

DVRPC's Land Use Vision is key to achieving the region's broad environmental and resource protection goals. The *Update: Connections 2050* Land Use Vision emphasizes Centers-based development and the preservation of agricultural and natural lands. The Land Use Vision divides the region into four typologies: Infill and Redevelopment areas, Emerging Growth areas, Rural Resource Lands, and the Greenspace Network (see Figure B-12). The overall goal of the Land Use Vision is to concentrate new growth with increasing density into Emerging Growth, Infill and Redevelopment areas, and Centers; while preserving the region's Rural Resource Lands and Greenspace Network. Doing so will protect sensitive natural resource areas, conserve productive agricultural lands, and create linear, connected greenspace corridors that mitigate flooding, improve water quality, and provide access to passive outdoor recreational opportunities for all.

Infill and Redevelopment Areas

Infill and redevelopment areas account for most of the region's existing development, occupying the full spectrum of land use typologies, from densely developed urban cores to first-

generation suburbs to low-density residential suburban subdivisions. They do not include scattered, isolated, or village-scale development in otherwise rural areas. Although these areas are largely developed, they offer a wide array of opportunities for redevelopment and infill development over the Plan's timeframe. Such opportunities include vacant parcels, underutilized parcels, parcels that can be repurposed for other uses, and opportunities to increase density in strategic locations.

Emerging Growth Areas

Emerging growth areas are typically undeveloped, greenfield areas in our region's suburban fringes that have been targeted by counties and local governments for new growth. Although they represent "new development," these areas are often near to one or more Plan Centers and should seek to extend the walkable and relatively more compact areas of those Centers.

Greenspace Network

The Land Use Vision aims to link and expand the region's existing protected natural areas into a Greenspace Network, joining together parks, forests, meadows, stream corridors, and floodplains in an interconnected system. The Greenspace Network is based on the twin principles of protecting core natural resource areas and linking them with greenways to create a connected system of naturally vegetated, open space spanning urban, suburban, and rural areas. The Greenspace Network is broken down into just over 100 corridors with the goal to permanently protect unprotected acres in the system through acquisitions, easements, and land use regulations. Each

corridor is named to promote its identity and brand it as a unique preservation project (see Figure B-13 and the corresponding list).

The Greenspace Network is a key for achieving regional high-priority environmental goals, including protecting and improving water quality, maintaining healthy ecosystems, providing access to natural areas and recreational opportunities, mitigating flooding, and building community resilience. The goal of the Greenspace Network is to permanently protect unprotected acres in the system through acquisitions, easements, and land use regulations.

Rural Resource Lands

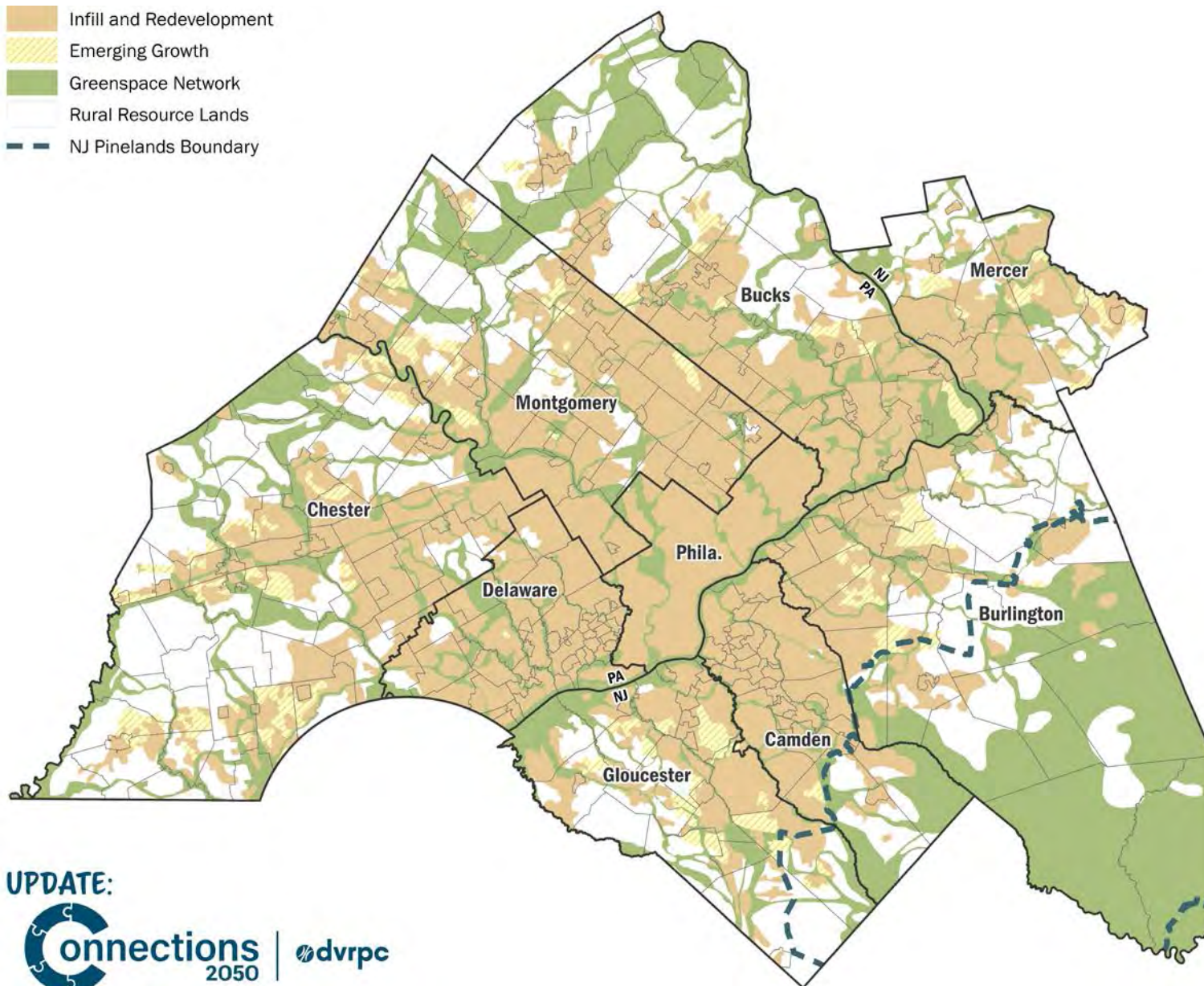
Rural Resource Lands are predominantly agricultural, natural, and rural areas worthy of heightened preservation efforts by governments and nonprofit land trusts. Rural Resource Lands comprise all the region's significant remaining agricultural areas, containing villages and scattered low-density development. Strategic acquisitions and easements, land use regulations, good stewardship, and appropriate forms of growth can maintain their integrity. Rural Resource Lands can accommodate limited growth that is in character with the local context.

Protecting these resources is critical to maintaining both the region's rural character and its farming economy. Although farming has always been a dominant economic sector in Greater Philadelphia, it has taken on new importance in recent years with the growing emphasis on locally produced food. There are

many advantages to locally produced food, including improved health, better food quality, and lower outlays of resources and materials for processing and transportation. Our region's highly productive, fertile soils—among the best in the Northeast—

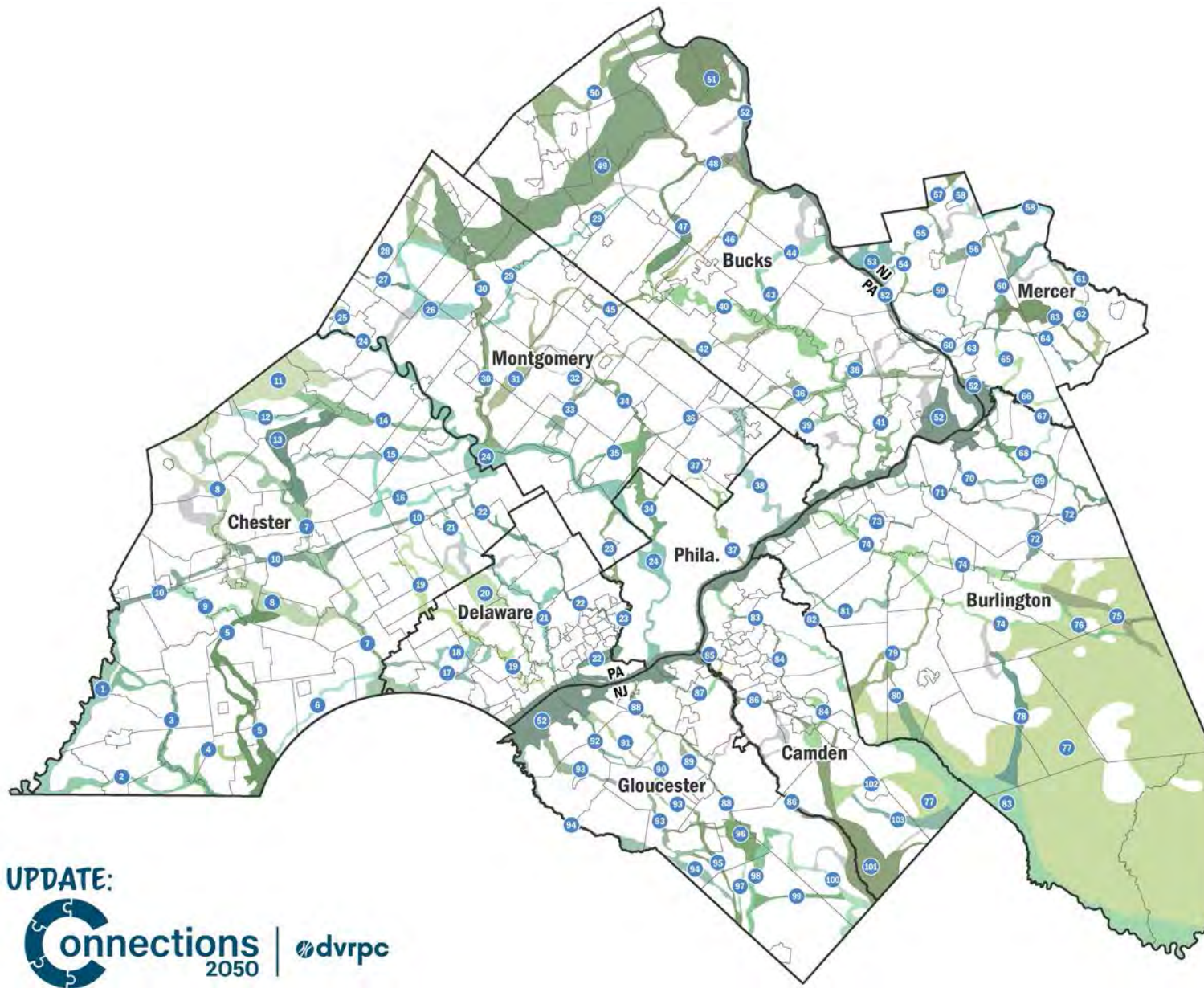
underpin the region's agricultural economy, but are permanently lost when farmland is converted to housing or other development.

Figure B-12: Land Use Vision



Sources: DVRPC, 2025; New Jersey Office of GIS, 2025.

Figure B-13: Greenspace Network



Source: DVRPC,
2025.

Greenspace Network Map Locations

- | | | | |
|------------------------------------|------------------------------------|--|---|
| 1. Octoraro Creek | 27. Minister Creek | 54. Jacobs Creek | 79. Southwest Branch
Rancocas Creek |
| 2. Serpentine Barrens | 28. Middle Creek | 55. Pennington Mountain | 80. Haynes Creek |
| 3. Big Elk Creek | 29. East Branch Perkiomen
Creek | 56. Stony Brook | 81. Pennsauken-Masons |
| 4. White Clay-Ways Run | 30. Perkiomen Creek | 57. North Hopewell | 82. South Pennsauken Creek |
| 5. White Clay Creek-Doe Run | 31. Skippack Creek | 58. North Mercer | 83. River to Bay |
| 6. Delaware Arc | 32. Towamencin Creek | 59. Shabakunk-Ewing | 84. Cooper River |
| 7. Brandywine Creek | 33. Stony Creek | 60. Delaware and Raritan
Canal | 85. Little Timber |
| 8. West Branch Brandywine
Creek | 34. Wissahickon Creek | 61. Millstone River | 86. Big Timber |
| 9. Buck Run | 35. Plymouth Meeting | 62. Big Bear Brook | 87. Woodbury Creek |
| 10. Great Valley Ridgelines | 36. Cross County Corridor | 63. Assunpink Creek | 88. Mantua Creek |
| 11. Big Wood Corridor | 37. Tacony-Cresheim Creek | 64. Miry Run | 89. Chestnut Branch |
| 12. Warwick-Elverson | 38. Pennypack Creek | 65. Pond Run-Back Creek | 90. Edwards Run |
| 13. Marsh Creek-Beaver Run | 39. Poquessing Creek | 66. Doctors Creek | 91. Repaupo Creek |
| 14. French Creek | 40. Neshaminy Creek | 67. Crosswicks Creek | 92. Pargy Creek |
| 15. Pickering Creek | 41. Mill-Queen Anne Creek | 68. Blacks Creek | 93. Raccoon Creek |
| 16. Valley Creek-Pigeon Run | 42. Little Neshaminy Creek | 69. Bacons Run | 94. Oldmans-Reed |
| 17. Harvey Run-Naaman's
Creek | 43. Mill Creek | 70. Crafts Creek | 95. Still Run (Maurice River) |
| 18. West Branch Chester
Creek | 44. New Hope-Ivyland | 71. Assicunk Creek -
Annaricken Brook | 96. Glassboro Wildlife
Management Area |
| 19. Chester Creek | 45. West Branch Neshaminy | 72. Budd Run-North Run | 97. Little Ease Run |
| 20. Ridley Creek | 46. Paunacussing-Pine Run | 73. Mill Creek | 98. Scotland Run |
| 21. Crum Creek | 47. Peace Valley-Deep Run
Creek | 74. Rancocas Creek | 99. Indian-Faraway |
| 22. Darby Creek | 48. Tohickon Creek | 75. Mount Misery | 100. Hospitality Branch |
| 23. Cobbs-Mill Creek | 49. North Woods | 76. Bishpams Mill Creek | 101. Great Egg Harbor River |
| 24. Schuylkill River | 50. Quakertown-Cooks Creek | 77. Pinelands Conservation
Areas | 102. Sleeper Branch |
| 25. Manatawny Creek | 51. Tinicum-Nockamixon | 78. Batsto-Friendship | |
| 26. Swamp-Deep Creek | 52. Delaware River | | |
| | 53. Washington Crossing | | |

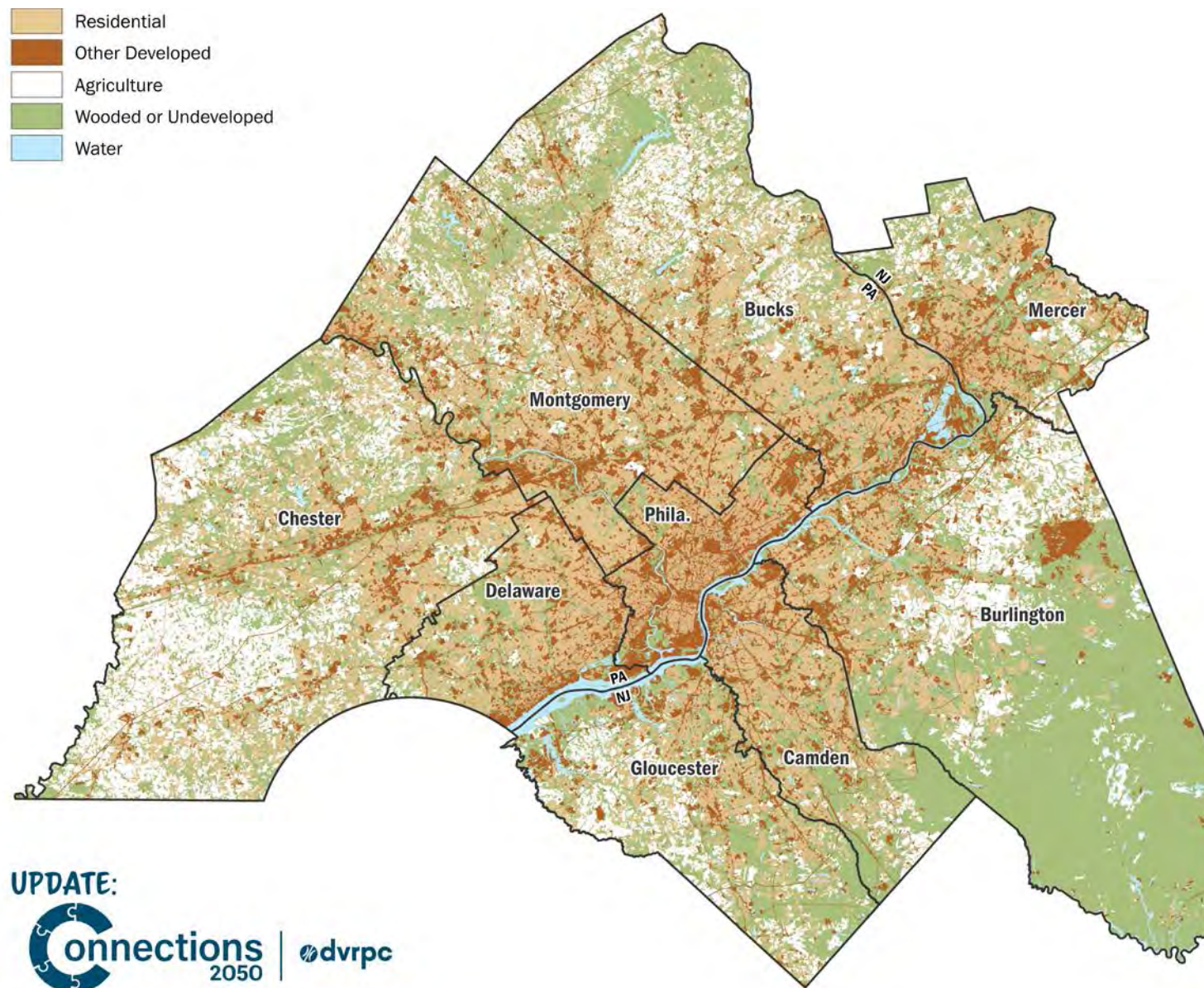
Open Space Inventory

DVRPC maintains an inventory of protected public and private open space to track the region's progress toward meeting its land preservation goals. The inventory tracks all publicly owned open space, preserved farmland, and privately-owned, protected open space. Between 2002 and 2024, the region has steadily increased its inventory of protected public and private open space. With over 644,400 acres of protected land to date, the region is nearly two-thirds of the way toward meeting the goal of permanently protecting one million acres of open space by 2040. Currently, 26 percent of Greater Philadelphia's total land area is protected open space (see Figure B-15 and Table B-7).

Land Use Mapping

DVRPC tracks land use in the region with an inventory that has been collected every five years since 1970 (excluding 1975 and 1985). DVRPC's *2020 and 2023 Enhanced Land Use Analytical Data Report* (ADR26109) dataset contains 73 different land use subcategories grouped under 13 primary categories, including residential, industrial, transportation, utility, commercial, institutional, military, recreational, agricultural, mining, wooded, undeveloped, and water (see Figure B-14).

Figure B-14: 2023 Land Use



Source:
DVRPC, 2023.

Figure B-15: Protected Open Space

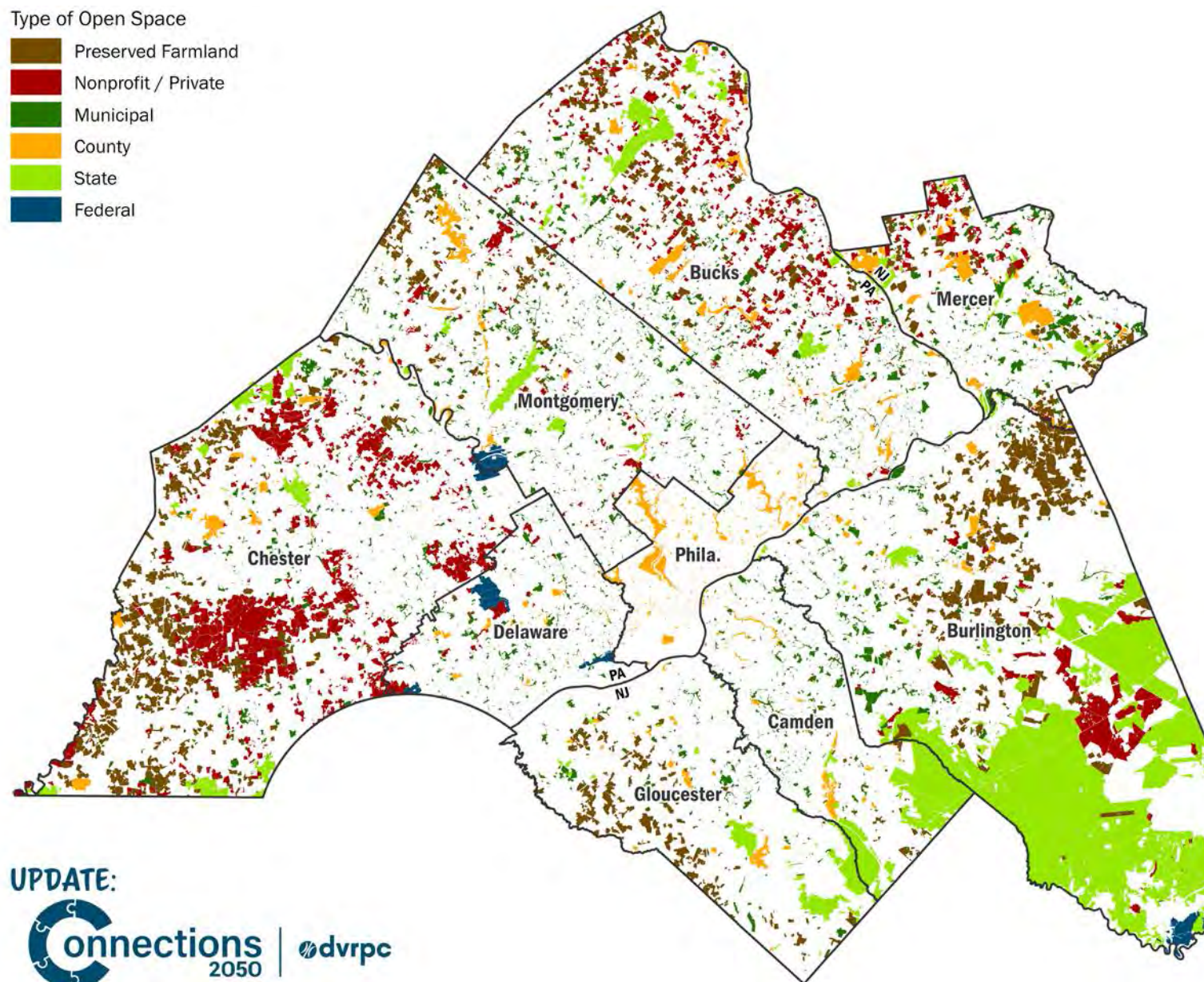


Table B-7: Protected Open Space Inventory, 2025

County	Public Protected Open Space					Private Protected Open Space			Total Protected	County Total Land Area	Percent of Land Area
	Federal	State	County	Municipal	Total Public	Nonprofit/Private	Preserved Farmland	Total Private			
Burlington	3,100.36	153,266.04	2,942.72	12,951.38	172,260.50	18,751.52	46,157.70	64,909.22	237,169.72	523,911.00	45.27%
Camden	0.00	20,666.24	3,044.26	5,701.25	29,411.75	6.84	2,837.74	2,844.58	32,256.33	145,673.00	22.14%
Gloucester	0.00	10,081.50	2,451.17	5,745.74	18,278.41	788.86	15,142.00	15,930.86	34,209.27	215,417.00	15.88%
Mercer	0.00	4,598.16	8,806.42	11,591.56	24,996.14	7,644.85	9,410.20	17,055.05	42,051.19	146,340.00	28.74%
NJ Subregion	3,100.36	188,611.94	17,244.57	35,989.93	244,946.80	27,192.07	73,547.64	100,739.71	345,686.51	1,031,341.00	33.52%
Bucks	0.00	12,865.05	9,370.56	17,860.38	40,095.99	27,719.81	19,707.80	47,427.61	87,523.60	397,831.00	22.00%
Chester	1,290.22	9,579.68	5,461.27	14,169.82	30,500.99	59,001.47	48,732.80	107,734.27	138,235.26	485,732.00	28.46%
Delaware	1,194.37	2,600.44	1,743.67	5,520.26	11,058.74	3,139.99	203.99	3,343.98	14,402.72	122,001.00	11.81%
Montgomery	2,155.55	4,979.56	5,272.37	15,937.83	28,345.31	6,381.65	10,759.49	17,141.14	45,486.45	311,737.00	14.59%
Philadelphia	372.44	300.82	11,718.81	0.00	12,392.07	673.55	0.00	673.55	13,065.62	91,178.00	14.33%
PA Subregion	5,012.58	30,325.55	33,566.68	53,488.29	122,393.10	96,916.47	79,404.08	176,320.55	298,713.65	1,010,648.00	21.21%
Regional Total	8,112.94	218,937.49	50,811.25	89,478.22	367,339.90	124,108.54	152,951.72	277,060.26	644,400.16	2,041,989.00	26.41%

All figures in acres. Figures may not add up due to rounding. Source: DVRPC, 2025.

Water Quality and Stormwater Runoff

In natural areas, such as forests, most rainfall soaks into the ground, where it is used by trees and other vegetation or is filtered through the soil to become groundwater. Only a small amount actually runs off land surfaces into waterways. In urban and built-up suburban areas, rooftops, streets, sidewalks, parking lots, and even compacted soils associated with lawns prevent rainwater from soaking into the ground. Instead, water drains off these impervious surfaces and is carried quickly by drains and pipes to rivers and streams. This stormwater runoff leads to non-point source pollution and increases the volume and velocity of stormwater, thereby eroding and enlarging stream channels and increasing sediment loads. The result is impaired water quality and degraded stream health.

It is important to effectively manage stormwater and improve water quality in urban and suburban settings. Techniques to manage stormwater in developed landscapes include:

- conservation landscaping – Uses native and climate-adapted plants, reduced mowing, and minimal chemical inputs to decrease runoff, improve soil health, and enhance habitat value;
- naturalized retention basins – Designed basins planted with native vegetation that temporarily store stormwater and allow it to slowly infiltrate or evaporate while filtering pollutants;
- turf reduction – Replacing traditional lawn areas with native plantings, meadow vegetation, or permeable

surfaces to reduce runoff and lower irrigation and maintenance needs;

- vegetated riparian buffers — managed zones of trees, shrubs and grasses planted alongside waterbodies to protect water quality, stabilize banks, and provide habitat; and
- engineered soil-vegetation systems commonly referred to as Green Stormwater Infrastructure (GSI), which soak up and slowly infiltrate stormwater by mimicking natural processes. GSI includes techniques such as rain gardens, green roofs, tree trenches, stormwater planters, and vegetated bioswales.

GSI also provides other benefits like improving air quality, greening the community, ameliorating the urban heat island effect, and fostering a sense of place.

Whether part of a sophisticated, engineered GSI approach or simply planted along a public right-of-way (ROW), street trees are one of the most effective forms of stormwater management and “greening” in an urban environment. One hundred mature tree crowns intercept approximately 100,000 gallons of rainfall per year. Translated into dollars, a single street tree produces \$90,000 of direct benefits, such as stormwater retention and air quality improvements, over its lifetime.

Historic Resources and Landscapes

Greater Philadelphia’s rich past is reflected in the variety and number of historic and cultural resources throughout the region. From indigenous archaeological sites to early Swedish

settlements, and from the colonial-era row houses of Society Hill to the hundreds of pre-20th century towns and villages that dot our landscape, Greater Philadelphia's history is incorporated into, and enriches the fabric of, present-day life. The wealth of resources is underscored by the number of historic landmarks, sites, buildings, and districts on the National Register of Historic Places, state and nationally recognized historic landscapes and heritage areas, sites protected through local historic designations, and countless other historic buildings and resources that lack any formal designation. These resources often form the bedrock of a community's character and identity, and are crucial in establishing the "sense of place" that is simultaneously a key ingredient and outcome of Centers-based development.

Despite sustained efforts by nonprofit organizations, government agencies, and local governments to identify, protect, preserve, rehabilitate, and restore the region's historic and cultural resources, these resources continue to be threatened by demolition, neglect, encroaching sprawl, incompatible land uses, poor planning, and insensitive design. These negative forces are often even more dangerous and destructive in Title VI communities that have no existing system of preservation and support, and fewer capital resources prohibit preservation and restoration efforts. The loss of these resources undermines key aspects of the Centers-based development philosophy, such as utilizing existing infrastructure, creating and celebrating a

community's unique character, protecting rural and agricultural landscapes, and enhancing human-scaled development patterns that promote walking and biking as viable transportation alternatives.

Transportation projects, in particular, can impair or destroy historic resources through road widenings, bridge replacements, realignments, and capacity enhancements, as has been experienced in communities across the region. In some instances, transportation assets, like bridges, are also historic. In these cases, targeted maintenance and care are needed to ensure their preservation. To ameliorate impacts on historic resources, federally funded transportation agencies must follow federal historic preservation laws and plan their projects accordingly. As part of this process, state historic preservation offices work with federal agencies to identify historic resources and avoid or minimize any potential adverse effects during the planning, permitting, design, and construction of federally funded and licensed projects. Since 2005, federal transportation regulations have established formal consultation requirements for MPOs and state DOTs to work with environmental, regulatory, and historic resource agencies in the development of long-range transportation plans. Additionally, DVRPC continually works with resource agencies and local governments to explore how transportation projects and local plans can better support, rather than impair, historic preservation and revitalization efforts.

